

Functional and Structural Characterization of Novel Protein **Mi-lipotrymase**, a
Mitochondrial-Associated Lipase-Protease from *Canis lupus*

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Abstract

After purifying a protein using techniques such as Molecular Sieve, Ion-Exchange, and affinity chromatography, a partial amino acid sequence had been determined. Initial analysis indicated that the protein is novel, and further research was necessary to determine its function and structure. The goal of this project was to conduct sequence analysis to predict and map the protein's functional and structural domains and motifs, sequence motifs, secondary structures, hydrophobic and hydrophilic regions, trans-membrane domains, and subcellular localization. Various bioinformatics tools were leveraged to predict the characteristics of the sequence. As a result of these queries, four major domains were identified: Smac_DIABLO, Lipase, PLAT, and Try_SPc. The secondary structure, as well as the hydrophobic and hydrophilic ranges in the sequence was also predicted. It was concluded that the protein does not contain a transmembrane domain and its subcellular localization is in the extracellular region and the mitochondria. Based on the combination of its domains, the protein appears to play a significant role in apoptosis, lipid metabolism, and protein degradation. Further 3D structure modeling and ligand-binding predictions helped support these proposed functions and provided insight into the molecular mechanism of action of this novel protein.

Objective

We have been given a novel protein purified by a combination of Molecular Sieve (Gel-Filtration), Ion-Exchange and affinity chromatography from HeLa cells with the following near-complete amino acid sequence:

MLRAALSTARRGPRLSRLLSAAATSAVPAPMAALKSWLSRSVTSFFRYRQCCLCVPVVAN
FKKRCFSELIRPWHKTVTIGFGVTLCAPVIAQKSEPHSLSSEALMRRVSLVTDSTSTFLSQ
TTYALIEAITEYTKAVYTLTSLYRQYTSLLGKMNSEEEDEVWQVIIGARAEMTSKHQEYL
KLETTWMTAVGLSEMAAEAAYQTGADQASITARNHIQLVKLQVEEVHQLSRKAETKLA
EAQIEELRQKTQEEGEERAESEQEAYLREDNEVCYERLGCFSDDSPWAGIVERPLKILPW
SPEKVNTRFLLYTNENPDNFAQEIVADPSTIQSSNFNTGRKTRFIIHGFIDKGEESWLSTMCQ
NMFKVESVNCICVDWKSGSRTAYSQASQNVRIVGAEVAYLVGVQLQSSFDYSPSNVHIIGH
SLGSHAAGEAGRRTNGAVGRITGLDPAEPCFQGTPELVRLDPSDAQFVDVIHTDIAPFIPN
LGFGMSQTAGHLDFFPNGGKEMPGCQKNVLSQIVDIDGIWQGT
RDFAACNHLRSYKYYTDSILNPDGFAGFSCASYSDFTANKCFPCSSEGCPQMGHYADRF
GRTKGVGQLFYLNTGDASN FARWRYRVDVTLSGKKVTGHVLVSLFGNKGNSRQYEIFQ
GTLKPDNTYSNEFDS DVEVG DLEKVKFIWYNNVINLTLPKVGASKITVERNDGSVFNFC
SEETVREDVLLTLTACVVGGEDARPHSWPWQISLQYLKNDTWRHTCGGTLIASNFLTA
AHCISNTRTYRVAVGKNNLEVEDEEGSLFVGVDTIHVHKRWNALLLRNDIALIKLAEHVE
LSDTIQVACLPEKDSLLPKDYPCYVTGWGRLWTNGPIADKLQQGLQPVDHATCSRIDW
WGRVVKKTMVCAGGDGVISACNGDSGGPLNCQLENGSWEVFGIVSFGSRRGCNTRKKP
VVYTRVSAYIDWINEK

The main objectives of this project are:

- 1) To predict the structure and function of the novel protein.
- 2) To identify the functional domains and motifs, sequence motifs, secondary structures, hydrophobic and hydrophilic regions, trans-membrane domains, subcellular localization, and protein sorting signals.
- 3) To predict a 3D structure of the domains and the overall structure of the protein.
- 4) To compare the correlations of this sequence to existing well-known protein sequences to find regions of similarities, identity, and domains.
- 5) To examine the possibilities of functional domains, membrane domains, hydrophobicity, secondary structures, molecular mechanisms, and identifying features.
- 6) To utilize all the aforementioned details in predicting disease(s) related to the relevant gene and predict future research in the field.

Background

Sequence analysis is a foundational component of bioinformatics. It can involve the examination of RNA, DNA, or protein sequences to uncover their underlying function, evolutionary relationships, and interactions with other molecules. Many computational sequence analysis algorithms for protein structures specifically exist to analyze and predict their structure and function and have drastically improved biological research. Specifically, tools such as BLAST, ClustalOmega, and PredictProtein allow bioinformaticians to sort through large amounts of biological data to predict and model protein structures quickly without the need for physical experiments. Wet-bench approaches, like gel-electrophoresis, Western Blotting, and cryo-electron microscopy, require physical samples, preparation, and long run times to execute them. While these methods provide direct and empirical evidence of protein presence and characterization, they are often resource intensive, time consuming, and limited in scale. For this project, we utilized computational analysis tools to analyze our novel protein's structure and function now that the sequence has been acquired. We have primarily conducted the research through sequence analysis-based strategies.

Different databases were searched to predict the properties of proteins such as its secondary structure, hydrophobicity, subcellular localization, sequence motifs, and binding sites. Most of these databases include data obtained from wet bench experiments while some databases make predictions using machine learning models and scoring systems such as PSSM and HMMs. The computing algorithms of the bioinformatics tools we used have been analyzed and peer-reviewed meaning they can be used to predict future unknown protein sequence functions and structures. With these methods, we can compare our novel protein to well known protein structures to analyze its components further in a short amount of time. However, our predictions will need to be validated through wet-lab procedures in the future to support its predicted structure and function.

Tools and Methods

1. Outline

Functional/structural domains and motifs

- a. **CDD (Conserved Domain Database):** CDD is a protein annotation resource that consists of numerous collections of annotated multiple sequence models for well-known protein domains, conserved sites, and proteins.
<https://www.ncbi.nlm.nih.gov/cdd/>¹
- b. **InterPro:** Interpro is a protein annotation database that consists of integrated protein family classifications. It classifies proteins into families by predicting domains, motifs, and conserved sites.
<https://www.ebi.ac.uk/interpro/>²
- c. **SMART (Simple Modular Architecture Research Tool):** SMART is a biological database that is used in the identification and analysis of protein domains, SMART uses profile-hidden Markov models to detect domains in sequences.
<http://smart.embl-heidelberg.de/>³
- d. **HMMER:** HMMER is a software suite developed based on Hidden Markov Models and is widely used for the identification of similar sequences.
<http://hmmer.org/>⁴

Sequence Motifs

- a. **NsitePred:** NsitePred is a server designed for sequence-based prediction of binding residues for five most populated nucleotides in PDB: ATP, ADP, AMP, GDP, and GTP.
<https://biomine.cs.vcu.edu/servers/NsitePred/>⁵
- b. **COACH:** COACH is a template-based algorithm for protein-ligand binding site prediction, estimated by SVM model.
<https://zhanggroup.org/COACH/>⁶
- c. **SWISS MODEL:** SWISS MODEL is an automated protein homology-modeling server that is designed to work with minimum user input, ie only the amino acid of a target protein.
<https://swissmodel.expasy.org/>⁷

Secondary Structure

- a. **PSIPred:** PSIPred is a popular bioinformatics web service that offers a wide range of machine learning based analyses for identifying protein structure and functions.⁸
<https://bio.tools/psipred>⁹

- b. **JPred4:** JPred4 is the latest version of well-known server JPred. It is widely used for protein secondary structure prediction.¹⁰
<https://www.compbio.dundee.ac.uk/jpred/>¹¹
- c. **PREDATOR:** PREDATOR is a secondary structure prediction method based on recognition of potentially hydrogen-bonded residues in the amino acid sequence.
https://npsa-prabi.ibcp.fr/cgi-bin/npsa_automat.pl?page=/NPSA/npsa_predator.html¹²
- d. **SCRATCH:** SCRATCH is a suite of tools used for predicting protein tertiary structure and other structural features including high sequence homology to known sequence.
<https://scratch.proteomics.ics.uci.edu/>¹³

Hydrophilic/Hydrophobic Regions

- a. **ePESTfind:** ePESTfinds objective PEST motifs within a protein sequence. These PEST motifs are located by ePESTfind by looking at stretches of at least twelve hydrophilic amino acids in length with a concentration of specific amino acids.
<https://www.bioinformatics.nl/cgi-bin/emboss/epestfind>⁴⁵
- b. **Expasy-ProtScale:** Expasy-ProtScale uses values from the Kyte & Doolittle hydropathy scale, where each amino acid is given a value based on their hydrophilicity or hydrophobicity.
<https://web.expasy.org/protscale/>⁴⁷
- c. **SCRATCH Protein Predictor- ACCPro:** SCRATCH is a suite of tools used for predicting protein tertiary structure and other structural features including high sequence homology to known sequence.
<https://scratch.proteomics.ics.uci.edu/>¹³

Trans-membrane domains

- a. **TOPCONS:** TOPCONS predicts the topology of a protein, specifically of its potential membrane spanning segments along with their orientation in or out relative to the membrane. The prediction is a consensus from five different topology prediction algorithms.
<https://topcons.cbr.su.se/>⁴⁹
- b. **DeepTMHMM-1.0:** DeepTMHMM is a deep learning algorithm that detects and predicts the topology of alpha helical and beta barrel proteins of any putative transmembrane proteins.
<https://services.healthtech.dtu.dk/services/DeepTMHMM-1.0/>⁵²
- c. **EMBOSS-TMAP:** EMBOSS-TMAP predicts transmembrane segments for an aligned set of protein sequences utilizing a multi algorithm approach, and presents results in a plot of the propensities in the amino acid sequence.
<https://www.bioinformatics.nl/cgi-bin/emboss/tmap>⁵⁰

Subcellular Localization / protein-sorting signals

- a. **BUSCA:** BUSCA (Bologna Unified Subcellular Component Annotator) is a

web-server that predicts the subcellular localization of proteins that integrates multiple tools to not only predict localization-related protein features, but also for detecting globular and membrane proteins.

<https://busca.biocomp.unibo.it/>⁵¹

- b. **WoLF PSORT:** WoLF PSORT predicts the subcellular localization of proteins based on their amino acid sequences, specifically by using sorting signal motifs and protein sequence features.
<https://wolfpsort.hgc.jp/>⁵³
- c. **ELSPred:** ELSPred predicts the subcellular localization of eukaryotic proteins by using physicochemical properties, amino acid composition, and dipeptide composition.
<https://webs.iitd.edu.in/raghava/eslpred/submit.html>⁵⁷
- d. **DeepLoc2.1:** DeepLoc 2.1 predicts the subcellular localization(s) of eukaryotic proteins and their sorting signal(s) using neural networking algorithms trained through using proteins located on Uniprot.
<https://services.healthtech.dtu.dk/services/DeepLoc-2.1/>⁵⁴
- e. **IPSort:** IPSort is a site for subcellular localization predictions using N-terminal sorting signals. It can predict both plant and non-plant sorting signals.
<https://ipsort.hgc.jp/>⁵⁵
- f. **TargetP-2.0:** TargetP-2.0 server predicts the presence of N-terminal presequence and a potential cleavage site.
<https://services.healthtech.dtu.dk/services/TargetP-2.0/>⁵⁹
- g. **CELLO:** CELLO is a subcellular localization predictor that uses the unique features of amino acid sequences to make its predictions.
<https://cello.life.nctu.edu.tw/>⁵⁶
- h. **HHpred:** HHPred is a tool used to predict protein homologies, function, and structure.
<https://toolkit.tuebingen.mpg.de/tools/hhpred>⁸⁴

Function Predictions

- a. **ProIdent:** ProIdent is a web-based server that identifies proteases and their types.
<http://www.csbio.sjtu.edu.cn/bioinf/Protease/>⁶⁶
- b. **DeepGOWeb:** DeepGOWeb is a web-based service that predicts protein function through a protein's amino acid sequence.
<https://deepgo.cbrc.kaust.edu.sa/deepgo/>⁶⁷
- c. **Interpro:** Interpro is a protein annotation database that consists of integrated protein family classifications. It classifies proteins into families by predicting domains, motifs, and conserved sites.
<https://www.ebi.ac.uk/interpro/>²

Principle of Algorithms

1. Functional/structural domains and motifs

- a. **CDD:** The Conserved Domain Database(CDD) is a free resource of collection of protein family and protein domain models. The database was developed and is managed by National Center for Biotechnology Information (NCBI). As of the latest release, March 2024, CDD contains more than 64,456 domain models, organized into 4,705 multi-model Superfamilies, obtained from various sources including Pfam, SMART, TIGERFAMS, and CDD's in-house data curation.¹⁴ CDD database contains models from various sources, which can often overlap in scope. The information from these redundant and homologous domain models is summarized at a superfamily level, and the domain annotation on proteins is flagged as either 'specific'(identifying molecular function with high confidence) or as 'non specific'(identifying superfamily membership only).¹⁵ CDD utilizes RPS-BLAST for rapid identification of conserved domains in protein sequences, using position-specific score matrices (PSSMs). Reverse Position-Specific BLAST (RPS-BLAST) compares protein sequences against a precompiled database of Position-Specific Scoring Matrices (PSSMs). This is a variation of PSI-BLAST and enables detection of sensitive conserved domains by aligning the query sequence to the precompiled PSSMs. RPS-BLAST does not build a new model, which enables it to be a rapid and sensitive algorithm. CDD assigns each conserved domain with an E-value. E-values give the number of sequences with a score more than the threshold that would be found by random chance. CDD's profile-based searching and default cutoff E-values makes the algorithm well-suited for the identification of well-known domains. CDD, hence provides precise domain hits and functional site annotations.¹⁶
- b. **InterPro:** InterPro is a freely accessible resource for classification of protein sequences into families, developed by European Bioinformatics Institute (EMBL-EBI). InterProt integrates multiple protein signature databases including CATH-Gene3D, the Conserved Domains Database (CDD), HAMAP, NCBIFAM, PANTHER, PIRSF, Pfam, PRINTS, PROSITE, the Structure-Function linkage Database (SFLD), SMART, and SUPERFAMILY into a single cohesive resource.¹⁷ These member databases signature are not automatically incorporated into InterPro; instead, they undergo a thorough manual inspection and integration process. Matches from several signature are evaluated with the recent version of UniProtKB to eliminate any false positives.¹⁷

InterPro uses InterProScan, a versatile software suite that runs the signature recognition methods associated with each member database. The algorithms employed include Hidden Markov Models (HMMs) for Pfam and SUPERFAMILY, regular expressions for PROSITE, fingerprint-based searches for PRINTS, and other database-specific techniques. Profile Hidden Markov Models are a way of turning a multiple sequences alignment into a position-specific scoring system, which is suitable for searching databases for remotely homologous sequences.¹⁸ InterProScan runs these tools in parallel, enabling comprehensive and efficient scanning of a protein sequence. HMMs, in particular, provide high sensitivity in detecting distant homologs by capturing position-specific variations within domain sequences. This multi-algorithmic framework allows InterPro to provide rich functional annotation and insight into protein structure and evolution.¹⁹

- c. **SMART:** SMART (a Simple Modular Architecture Research Tool) is a widely used web based tool, maintained by the European Molecular Biology Laboratory(EMBL), that allows the identification and annotation of genetically mobile domains and the the analysis of domain architecture. SMART was first introduced by Schulz *et al* as a database for signaling domains, where it aimed to address challenges in understanding the evolution and function of multi-domain proteins, particularly those with highly divergent sequences. Post expanding the domain coverage, SMART adopted a relational database system.²⁰ SMART uses profile Hidden Markov Models to detects domains from sequences with relatively high selectivity and specificity. SMART also employs per-species protein clustering which helps against potential biological roles even to previously novel proteins. Once categorized into a species-specific database, all proteins are subsequently grouped into clusters using the CD-HIT algorithm, applying a 96% sequence identity threshold to ensure high similarity within each cluster. SMART merges several features such as signal sequences, transmembrane helices, coiled coil regions, and compositionally biased regions into a single line output.²¹ The following priority list is used to resolve overlapping predictions based on the perceived prediction accuracy: Domain > Signal > TM > Coils > Seg. SMART also uses E-value for the static significance. SMART, therefore, can be effectively used for predicting functional and structural domains and motifs.²²
- d. **HMMER:** HMMER is a fast, free to use, and sensitive web server used for homology searches. The database enables sensitive sequence analysis by using profile Hidden Markov Models. HMMER allows for users to summarize the resulting sequence in multiple ways—users can view and filter the significant hits by domain architecture or taxonomy.²³ It is particularly valuable for identifying

protein domains, gene families, and evolutionary relationships that might otherwise be undetectable due to low sequence identity. HMMER has introduced multiple search models including `hmmsearch`, `hmmsearch`, `phmmer`, and `jackhmmer`. Each of these are tailored for different types of query configurations.²⁴ HMMER integrates accelerated search algorithms using vector instructions and probabilistic filtering to ensure computational efficiency. When a query sequence is submitted, a query HMM is generated by the software. This query HMM is then used for searching against the database. For each matching query, statistical significance is given by two e-values: conditional e-values and individual e-values. The individual E-values represents the significance of a domain hit as if it were the only one detected in the sequence. On the contrary, the conditional E-value estimates the statistical significance of a domain assuming it is a genuine homolog.²⁵ HMMER serves as a powerful engine for detecting functional and structural domains and motifs.

Comparison of Functional/Structural Domains and Motifs Prediction Tools:

While CDD, InterPro, SMART, and HMMER all specialize in the identification of functional and structural domains and motifs, each tool has a unique approach and different algorithms and integration methods. CDD relies on RPS-BLAST, which offers fast identification of conserved domains through comparison against a precompiled library of PSSMs. SMART and HMMER both utilize profile Hidden Markov Models (HMMs), which provide a high level of sensitivity by capturing position-specific variation in domain alignments. However, HMMER allows more flexibility through search modes like `hmmsearch` and `phmmer`, and provides two types of E-values (individual and conditional), adding an extra layer of statistical context to the predictions. InterPro integrates signatures from multiple member databases such as Pfam, PROSITE, and SUPERFAMILY. It uses InterProScan to run diverse algorithms in parallel, increasing coverage and reducing false positives through manual curation. In our view, CDD and HMMER are the best databases because they use specific scoring systems, and have diverse E-value calculations. If two or more scoring systems give similar output, then we can verify the result.

2. Sequence Motifs

- a. **NsitePred:** NsitePred is an online server that provides an ensemble of predictors of binding residues for five common nucleotides in the PDB including ATP, ADP,

AMP, GTP, and GDP. It performs especially well for the sites where residues are clustered together in a sequence.²⁶ The model can also reveal several sequence-derived hallmarks of the nucleotide-binding residues, including high residues-level conservation and specific patterns of secondary structure and amino acid pair arrangements in their surrounding regions.

NsitePred implements a consensus of machine learning based methods and alignment-based predictors. It utilizes PSIPRED for secondary structure prediction, REAL-Spine3 for solvent accessibility and dihedral angles, and PSI-BLAST to generate PSSM profiles. These features are processed using a sliding window and classified by a support vector machine(SVM) model using a one-against-the-rest strategy (SVMpred). NsitePred then combines both predictions by averaging their outputs to generate a final consensus prediction.²⁶

- b. **SWISS-MODEL Expasy:** SWISS-MODEL is a server for automated modeling of the dimensional (3D) protein structures. The server has three different interaction modes. The ‘first approach mode’ provides a simple interface and requires an amino acid sequence input. The ‘alignment mode’ allows the user to specify which sequence in the given alignment is the target sequence and which one corresponds to a known protein chain on ExpASy (Expert Protein Analysis System). Lastly, the ‘Project mode’ allows users to manually optimize modeling requests. While the server is primarily used for 3D modeling, it also provides valuable insights into functional annotations such as ligand-binding sites, and metal coordination which can often be inferred through structural alignment and annotated templates.

SWISS -MODEL averages the backbone atom positions of the template to generate the core of the model. The templates are then weighed by their sequence similarity to the target sequence. To generate the missing atom positions, an ensemble of fragments compatible with the neighboring stems is constructed using constraints space programming (CSP). This modeling process includes template identification via BLAST and HHblits, target-template alignment, and model building using the ProMod3 engine. The structural conservation criteria includes ligand and cofactor annotations, which allows users to highlight potential sequence motifs involved in ligand binding.²⁷

- c. **COACH:** Comparison of Alignments by Constructing Hidden Markov Models (HMMs) is one of the most efficient approaches to protein-ligand binding sites prediction. COACH utilizes molecular docking to refine the ligand-binding poses. The server requires either an amino acid sequence or a three-dimensional structure of the query protein as an input. What makes COACH effective for sequence motifs discovery is its ability to extract conserved patterns of residues

involved in ligand binding by comparing the input protein to a large set of experimentally annotated protein-ligand complexes from the BioLiP database. Once COACH models the protein structure of the protein via the I-TASSER Suite, it submits the structure to five individual methods—COFACTOR²⁸, FINDSITE²⁹, TM-SITE³⁰, ConCavity, and S-SITE³¹—to predict the protein-ligand binding sites, by matching them with templates from BioLiP³². The results from the aforementioned individual methods are then combined to consensus predictions by the COACH algorithm³³, which highlights the most statistically significant and structurally conserved motifs. COACH uses confidence values, known as C-scores, ranging from 0 to 1 as a statistical significance approach. C-scores are assigned for each recognized ligand binding site. This ensemble-based strategy makes COACH a powerful tool for identifying motifs that are crucial to ligand recognition and binding.

Comparison of Sequence Motifs Prediction Tools:

NsitePred, SWISS-MODEL, and COACH all serve the purpose of identifying sequence motifs, they each adopt distinct strategies and strengths depending on the input data and target application. NsitePred is highly effective for detecting nucleotide-binding motifs and performs best when the binding residues are spatially clustered within the sequence. It combines machine learning techniques (SVM-based SVMpred) with alignment-based approaches. In contrast, SWISS-MODEL is primarily a homology modeling tool but extends its utility by leveraging structural alignments to infer ligand-binding motifs through conserved residues and template annotations. Its use of constraint space programming (CSP) and the ProMod3 engine enables fine-tuned modeling, where motifs involved in ligand recognition can be inferred indirectly from high-quality 3D models. COACH, on the other hand, combines multiple ligand-binding prediction tools and integrates molecular docking with template-based structural alignment. Its strength lies in matching conserved ligand-binding patterns from the BioLiP database using Hidden Markov Models (HMMs) and generating C-scores to express statistical confidence. While NsitePred focuses more on sequence-derived features and secondary structures, COACH and SWISS-MODEL capitalize on 3D structural alignments and template conservation. Using them in combination can enhance the accuracy of motif prediction, especially when sequence-based and structure-based methods converge on similar binding patterns.

3. Secondary Structures

- a. **PSIPred:** PSIPred is a highly accurate secondary structure prediction server that allows users to perform prediction from amino acid sequences. PSIPred combines neural networks with position-specific scoring matrices (PSSMs) generated by PSI-BLAST. Neural network is a machine learning approach that makes devices in a similar manner to that of human brain, by minimizing the way biological neurons work together.³⁴

This method allows the server to precisely predict the likelihood of each residue forming an alpha-helix, beta-strand, or coil, and achieving prediction rates of around 80%.

The PSIPred algorithm begins with running PSI-BLAST to identify homologs by aligning the input sequence against a protein database and creating a PSSM. One advantage of using PSI-BLAST is that it is far less time consuming than for other servers.³⁵ The result is then fed into a four-feed-forward neural network that assigns probabilities to each secondary structure type. The server uses confidence level as a statistical significance approach. The confidence level is assigned to each amino acid, which is graphically visualized in the result as a heatmap. The output hence, is a residue by residue annotation of secondary structure, which can be a value tool for further tertiary structure modeling, binding site prediction, and motif recognition.³⁶

- b. **JPred4:** JPred4 is the most recent version of the popular JPred protein secondary structure prediction server. The server provides predictions by the JNet algorithm. JPred4 features high accuracy, with blind three-state (alpha-helix, beta-strand, and coil) secondary structure prediction accuracy of 82.0%.³⁸ The users are able to submit a single protein sequence, or multiple sequence alignment(MSA) for prediction.

The neural network based predictor, JPred3, was retrained to JNet 2.3.1 using 7-fold cross-validation on 1358 SCOPe/ASTRAL superfamily sequences, and the multiple sequence alignment are generated via PSI-BLAST against UniRef90. Cross-validation is a technique for evaluating ML models by training several ML models on subset of the available input data and further evaluating them on the complementary subset of the same data.³⁷ JPred server provides a rich end-user experience while providing significantly accurate protein secondary structures.³⁸

- c. **PREDATOR:** PREDATOR is a secondary structure prediction tool that identifies potential hydrogen-bonded residues within an amino acid sequence to classify them into different secondary structure types. It begins by aligning the input sequence to homologous sequences with known secondary structures and then applies the DSSP method to assign structural codes based on hydrogen bonding

patterns. Dictionary of Secondary Structure of Proteins (DSSP) provides an elaborate description of the secondary structure elements in a protein structure, including backbone hydrogen bonding. These patterns are further processed using machine learning approaches, including neural networks, to predict the final secondary structure.³⁹

The tool provides a color-coded output for easy interpretation and typically achieves an accuracy between 70–75%. For statistical significance, PREDATOR assigns a reliability score ranging from 0 to 1 for each residue, which is visualized as a graph in the final prediction output, helping users gauge the confidence of individual residue-level predictions.⁴⁰

- d. **SCRATCH:** SCRATCH is a web-based server designed for predicting various structural features of proteins, including secondary and tertiary structures, relative solvent accessibility, domains, and disulfide bridges. Developed using a combination of machine learning algorithms, evolutionary profiles, PDB-derived fragment libraries, and energy functions, SCRATCH offers a robust platform for structure-based protein analysis. Its predictions are especially useful for proteins without known homologs.⁴¹

For secondary structure prediction, SCRATCH uses two specialized modules: SSPro and SSpro8. SSPro predicts protein structures in three broad categories: alpha-helix, beta-strand, and coil. SCRATCH leverages PSI-BLAST profiles, the PDB training set, and 1D recurrent neural networks (RNNs). Although SCRATCH does not explicitly provide statistical significance scores (such as E-values or confidence scores) for individual predictions, it demonstrates strong performance, with SSPro achieving over 84% accuracy and SSpro8 reaching about 72% accuracy on sequences lacking homologs in the PDB. This performance makes SCRATCH a highly effective tool for both functional annotation and structural prediction in computational biology workflows.⁴¹

Comparison of Secondary Structure Prediction Tools:

While PSIPred, JPred4, PREDATOR, and SCRATCH all predict secondary structures, they differ in algorithms, accuracy, and how they report confidence. PSIPred uses PSI-BLAST and a neural network to classify residues into helix, strand, or coil, achieving around 80% accuracy and offering confidence scores visualized as a heatmap. JPred4, based on the JNet algorithm, improves slightly with 82% accuracy through retraining with 7-fold cross-validation on a curated superfamily dataset and supports both single sequences and MSAs as input. PREDATOR uniquely focuses on hydrogen bonding patterns using the DSSP method and neural networks, and although slightly less accurate (70–75%), it

provides per-residue reliability scores ranging from 0 to 1, making it helpful for gauging prediction confidence. SCRATCH uses recurrent neural networks and PSI-BLAST profiles, reaching up to 84% accuracy with SSPro, especially effective on proteins without known homologs, although it does not report individual confidence values. Each tool offers different strengths. PSIPred and JPred4 excel in statistical interpretation and accuracy, PREDATOR adds unique insight through H-bond analysis, and SCRATCH provides comprehensive predictions beyond just secondary structure. Using a combination of these tools can increase reliability and provide a more complete structural overview.

4. Hydrophilic/Hydrophobic Regions

- a. **ePESTfind:** ePESTfind is a bioinformatics program used to identify PEST motifs in protein sequences. These motifs were first described by Rogers et al. in 1986 as regions rich in proline (P), glutamic acid (E), serine (S), and threonine (T). Positively charged amino acid residues were noted to often flank these PEST motifs and were also discovered to be associated with being regions of rapid proteolytic degradation.

The ePESTfind algorithm searches for regions in a protein sequence that are flanked by arginine, histidine, or lysine, through a sliding window with a default length of 10 amino acids. A sequence is considered a PEST motif if it includes at least one Proline, one aspartate or glutamate, and one serine or threonine. Sequences that lack these components are excluded from scoring. Valid motifs are then evaluated using the Kyte-Doolittle hydrophobicity scale. PEST scores can range from -45 to +50, with scores above +5.0 considered potential PEST degradation signals. However, scores below +5.0 are viewed as having poor potential as a PEST motif and are deemed biologically insignificant. The scoring threshold is empirically derived, and changes are discouraged without strong biological justification. ePESTfind is unique due to its requirement for positively charged flanking residues, which boosts specificity. In addition, hard-coded hydropathy values from the Kyte-Doolittle hydropathy scale also contribute to consistent and reproducible scoring.⁴⁵

- b. **ExPASy-ProtScale:** ProtScale is a web-based program available on ExPASy, which generates various types of profiles of protein sequences in a graphical format through the use of various scales. These scales contain numerical values that quantify amino acids' different physical or chemical properties, such as hydrophobicity. ProtScale has a wide range of scales, including 50 predefined

scales derived from literature, including widely used ones like the Kyte-Doolittle hydrophathy scale, which was used in this project.

The algorithm operates through a sliding window that can be customized. At each position, the average value of amino acids within the window is calculated and assigned to the central residue. Users can customize the weighting method depending on their goals. This allows users to adapt analyses based on the property of amino acids being studied. The results are shown in a line-graph format which emphasizes the variation across the sequence. ProtScale does not provide statistical significance values. Rather, users interpret the results visually or they can compare it across different graphs made from various scales. A unique strength of ProtScale is its broad applicability and customization options, which make it suitable for a multi-purpose protein property analysis.⁴⁷

- c. **SCRATCH Protein Predictor-ACCPro:** SCRATCH is a program used for predicting protein tertiary structures and structural features, like hydrophobicity. SCRATCH combines machine learning methods and evolutionary libraries from PDB to predict overall protein structure. The program includes 16 total modules as a method used for each prediction type, one of them being ACCPro. ACCPro predicts how exposed or buried each amino acid is in a protein, labeling its hydrophilic, denoted as (e) and hydrophobic regions, denoted as (-). ACCPro uses neural networks, and multiple sequence alignments created by PSI-BLAST to mark each residue as exposed or buried based on its accessibility while using a set threshold of 25% accessible.⁴¹

Comparisons of Hydrophilic/Hydrophobic Regions Prediction Tools:

All of these tools are great resources for predicting the hydrophobicity of a novel protein, but each has their strengths and weaknesses. While Expasy ProtScale helps at examining various amino acid properties it does not include statistical confidence, limiting its value. ePESTfind is good at detecting PEST motifs, but relies on a fixed scoring system without statistical validation. While ProtScale and ePESTfind are valuable for their specific and respective analyses, they lack the adaptability and predictive strength, unlike SCRATCH-ACCpro. At ACCPro's threshold of 25%, if exposed and buried residues are about equal in number, then ACCpro accurately predicts about 81% of residues present. Accuracy can increase to 90% if similar protein structures are already known in PDB.⁴⁸ This tool is the best in terms of predictive accuracy for hydrophobicity predictions, since it uses neural networks and evolutionary data to classify residue accessibility with up to 90% accuracy.

5. Trans-membrance domains

- a. **TOPCONS:** TOPCONS is a web-based server that uses other web-based membrane topology and signal peptides to make predictions using amino acid sequences. Specifically, it combines predictions based on five web-centered methods, including OCTOPUS, Philius, PolyPhobius, SCAMPI, and SPOCTOPUS. TOPCONS can classify parts of a sequence as belonging to a signal peptide (S), membrane region (M), or being on the inside (i) or outside (o) of a membrane.
Data is processed by a dynamic programming algorithm structured around Hidden Markov Models (HMMs) using the four-character alphabet (S, M, i, and o). The final topology prediction is generated by identifying the highest-scoring path through this model using a Viterbi-like algorithm. A ΔG -scale range, or the free energy scale, is also calculated and used to predict the propensity of each amino acid to be found within a transmembrane region of a membrane protein. Within the HMM, a reliability score is given.⁴⁹
- b. **DeepTMHMM-1.0:** DeepTMHMM is based on a deep learning model that takes a protein sequence as an input and determines the topology and type predictions by utilizing a state space model. The type predictions are denoted as signal peptide (S), inside cell/cytosol (I), alpha membrane (M), beta membrane (B), periplasm (P), and outside cell/lumen of ER/Golgi/lysosomes (O).
Five types of proteins were used for the training and testing of DeepTMHMM. They included alpha-helical transmembrane proteins with and without a signal peptide, beta-barrel transmembrane proteins, and globular proteins with and without a signal peptide. A very limited amount of eukaryotic beta-barrel structures were available to be used to train the model, so DeepTMHMM is primarily set up to predict the topology of prokaryotic beta barrels regarding queries related to that. However, small-scale testing on the limited eukaryotic beta-barrel transmembrane structural data available indicates that DeepTMHMM can correctly identify them and capture the main features of their topology.⁵²
- c. **EMBOSS-TMAP:** EMBOSS-TMAP predicts transmembrane segments in proteins through multiple sequence alignments and transmembrane propensity scoring. Therefore, the method relies on conserved sequence features to make these predictions but does not provide statistical confidence values. The web-based tool also analyzes peptide segments based on hydrophobic segments and terminal regions, and can also identify helices between 15 and 29 residues in length.⁵²

Comparisons of Trans-membrane Domain Prediction Tools:

In this project, TOPCONS is the best method for predicting transmembrane domains. The tool integrates several prediction methods and provides consensus results using multiple tools, reliability scores, and statistical confidence in its predictions. While TMAP uses information based on conserved sequence features, which are good for finding evolutionary relationships, and multiple

alignments, TOPCONS is more reliable due to its use of various methods along with the availability of confidence scores to its predictions. Regarding topology prediction accuracy, DeepTMHMM performs slightly worse than TOPCONS29 and slightly better than other transmembrane prediction tools, like CCTOP10 and DMCTOP11. Therefore, while DeepTMHMM uniquely uses a deep learning sequence-to-sequence model, TOPCON still outperforms it by a slight margin.⁵² Therefore, among the three, TOPCONS is the best choice.

6. Subcellular Localization / Protein-Sorting Signals

- a. **BUSCA:** BUSCA is a web-based tool used to predict the subcellular localization of proteins through their amino acid sequence. The tool uses a combination of numerous machine learning and deep learning strategies to make its predictions and was created by the Bologna Biocomputing Group. The tools detect characteristics such as signal peptides, transit peptides, GPI anchors, and both alpha-helical and beta-barrel transmembrane domains.
To do this, BUSCA uses DeepSig to detect secretory signal peptides, TPpred3 for mitochondrial and chloroplast signal detection, and PredGPI for GPI-anchoring site location. Finally, ENSEMBLE3.0 and BetAware are used to find alpha-helical and beta-strand membrane regions, respectively. It uses scoring measures such as the C-score, which represents the coverage of the sequence used for which BUSCA can make a prediction. Combining multiple specialized tools into a single platform to predict protein subcellular localization is one unique characteristic that allows the tool to be reliable.⁵¹
- b. **WoLF PSORT:** WoLF PSORT predicts subcellular compartments of proteins. It makes these predictions by using a simple k-nearest neighbor algorithm, which compares the query protein to sequences with known subcellular localization. After the input, results on WoLF PSORT include a list of analogous proteins along with tables that include specific features used in the prediction. Sequence alignments and links to UniProt and Gene Ontology are also present in the table which provides more context as to why a certain prediction was made. Most data is sourced from UniProt version 45 and incorporates localization information from Gene Ontology. WoLF PSORT has about 70% accuracy when predicting common locations like the nucleus, mitochondria, cytosol, plasma membrane, extracellular space, and chloroplast (in plants). For other areas like the Golgi or peroxisome, accuracy is lower.⁵³
- c. **ESLPred:** ESLPred is a tool used to predict the localization of eukaryotic proteins using a Support Vector Machine (SVM)-based method. The approach utilizes three types of protein characteristics which include physicochemical properties, amino acid composition, and dipeptide composition, which allows the tool to predict accuracies of 77.8%, 78.1%, and 82.4%, respectively. Additionally, data from PSI-BLAST was integrated to increase its prediction reliability. A

hybrid model was created by integrating PSI-BLAST and SVM-based methods, and prediction accuracy was increased significantly to 88.0%. The training dataset used in the creation of ESLPred came from version 33.0 of SWISS-PROT and includes only complete proteins with experimentally verified subcellular locations. The dataset was filtered to remove proteins with over 90% sequence similarity to reduce redundancy.⁵⁷

- d. **DeepLoc2.1:** DeepLoc2.1 was created by DTU Health Tech, which is a research institute based in Denmark. The web-based tool provides predictions of a protein's subcellular localization and sorting signals and also classifies the input protein sequence into the different membrane protein types which include transmembrane, peripheral, lipid-anchored and soluble. It uses neural network algorithms that were trained based on Uniprot protein entries along with experimental evidence of subcellular localizations for different eukaryotic proteins, allowing for accuracy of up to 93%. The output not only shows the predicted subcellular localization and/or signal peptide, but also exactly where it found them in the sequence. Certainty levels of the predicted label are also present and are shown in different shades of green, where the darker the green, the more certain the prediction based on the relative threshold.⁵⁴

- e. **iPSORT:** iPSORT is a subcellular localization web-based predictor that looks for N-terminal sorting signals. It has the ability to predict if a protein sequence contains a Signal Peptide (SP), Mitochondrial Targeting Peptide (mTP), or Chloroplast Transit Peptide (cTP). The algorithm uses a rule-based system, specifically by utilizing a decision list with three different nodes, each important for determining the protein's subcellular localization based on the first 30 amino acids. The first node checks for signal peptide (SP) patterns. If it does not pass, the second node examines if the sequence has a mitochondrial targeting peptide (mTP) or chloroplast transit peptide (cTP). If neither, then the sequence is classified as "other". The final node is used only for plants to further distinguish the sequence and cTP or mTP patterns. Each node uses amino acid indexing to examine the overall composition of the sequence, specifically looking at the first 30 residues' hydropathy index, total negative charge, and isoelectric point. Alphabet indexing is also used to classify each residue as numbers 0-2, where pattern indexing is then used to look for any recurring number patterns. These patterns are sourced from databases like the AAIndex Database. The output displays confidence intervals and hits seen in the query, making it a useful tool for detecting peptide signals.⁵⁵

- f. **TargetP 2.0:** TargetP 2.0 was created by DTU Health Tech, which is a research institute based in Denmark. It is a web-based method for finding sorting signals at the N-terminal. TargetP 2.0 detects proteins found in secretory pathways, mitochondria, and chloroplasts through a machine learning algorithm. Specifically, the tool is based on a deep learning algorithm using long short-term memory (LSTM) neural networks. TargetP 2.0 has been trained using a dataset of

protein sequences that have known subcellular localizations and sorting signals. The output of a TargetP2.0 not only shows the predicted subcellular localization and/or signal peptide, but also exactly where it found them in the sequence. Certainty levels of the predicted label are also present and are shown in different shades of green, where the darker the green, the more certain the prediction based on the relative threshold. Compared to other tools, TargetP 2.0 predicts ~97% of the SPs correctly.⁵⁹

- g. **CELLO:** CELLO uses a web-based server that predicts subcellular localization of proteins using dipeptide and amino acid composition, along with the physicochemical properties of the sequence. It was first developed at National Chiao Tung University in Taiwan. The algorithm uses a SVM (Support Vector Machine) algorithm that uses a two level classification system, with three nodes at each level. Sequence information gets passed through each node on both levels. After, the information is combined into one final prediction. The classification system does not rely on homology searches, allowing it to be relatively unaffected by homology of the sequences. Compared to other tools, CELLO is reported to predict eukaryotic sequences at an accuracy of around 87%.⁵⁶
- h. **HHPred:** HHpred is a web-based that is used to detect protein homologies along with structures that are remote. To do this, HHpred uses pairwise sequence alignments along with Hidden Markov Models (HMMs). HHpred also culls data from databases such as SCOP, PDB, Pfam, CDD, SMART, and COGs. HHpred then compares the query's HMM profile against data in the databases. This allows HHpred to detect distant evolutionary relationships. Results that are returned include multiple statistical scores. First, the probability value shows the likelihood that the match is a true homolog. Typically, percentages above 90% are generally considered to be valued as high confidence levels. The E-value, or expectation value, predicts how often a match might appear due to chance, with lower values indicating more statistically significant matches. The Score value represents the raw alignment score between the query and HMMs. Regarding this, higher score values usually indicate better matches. Finally, the SS (Secondary Structure) score measures how well the predicted secondary structure of the query aligns with the known structure of the template.⁸⁴

Comparisons of Subcellular Localization / Protein-Sorting Signal Prediction

Tools Compared to BUSCA and WoLF PSORT, which also predict subcellular localization through various algorithms for machine learning techniques, this SVM-based method seems the best. Though BUSCA integrates several methods, and WoLF PSORT uses k-nearest neighbor classification based on sorting signals and amino acid composition, the hybrid SVM model in this study achieves a high accuracy of 88.0%, making it a strong alternative, particularly when PSI-BLAST information is included. That said, DeepLoc2.1 and TargetP2.0 are two of the best

tools for subcellular localization and protein-sorting signal prediction tools. Not only do they both have higher potential for accuracy, but also provide a wide range of capabilities. Moreover, the use of neural networks trained on previous scientifically proven protein localizations makes them valuable tools. HHPred is a unique tool that is useful for identifying potential cell localizations along with homologous sequences and structures. It is also useful for predicting the organism that a protein may be found in.

7. Function Predictions

- a. **DeepGOWeb:** DeepGOPlus is a web-based protein function prediction tool that uses neural networks along with sequence similarity searches based on information in Gene Ontology.
It first detects important patterns and motifs directly from the protein's amino acid sequence, it then compares them to known proteins. DeepGOWeb is the online platform that allows users to easily access. When a user submits a protein sequence, DeepGOWeb predicts multiple potential functions and links to Gene Ontology (GO) terms. Results can describe predicted biological processes, cellular components, and molecular functions. Each of these predictions is assigned a confidence score between 0.1 (low certainty) and 1.0 (high certainty). The website allows users to adjust the confidence threshold to filter out weaker predictions.⁶⁷
- b. **Interpro:** InterPro is a freely accessible resource for classification of protein sequences into families, developed by European Bioinformatics Institute (EMBL-EBI). InterProt integrates multiple protein signature databases including CATH-Gene3D, the Conserved Domains Database (CDD), HAMAP, NCBIFAM, PANTHER, PIRSF, Pfam, PRINTS, PROSITE, the Structure-Function linkage Database (SFLD), SMART, and SUPERFAMILY into a single cohesive resource.¹⁷ These member databases signature are not automatically incorporated into InterPro; instead, they undergo a thorough manual inspection and integration process. Matches from several signature are evaluated with the recent version of UniProtKB to eliminate any false positives.¹⁷
InterPro uses InterProScan, a versatile software suite that runs the signature recognition methods associated with each member database. The algorithms employed include Hidden Markov Models (HMMs) for Pfam and SUPERFAMILY, regular expressions for PROSITE, fingerprint-based searches for PRINTS, and other database-specific techniques. Profile Hidden Markov Models are a way of turning a multiple sequences alignment into a position-specific scoring system, which is suitable for searching databases for remotely homologous sequences.¹⁸ InterProScan runs these tools in parallel, enabling comprehensive and efficient scanning of a protein sequence. HMMs, in

particular, provide high sensitivity in detecting distant homologs by capturing position-specific variations within domain sequences. This multi-algorithmic framework allows InterPro to provide rich functional annotation and insight into protein structure and evolution.¹⁹

- c. **ProtIdent:** ProtIdent is a free web-based tool used to predict proteases along with the six categories of them, including aspartic, cysteine, glutamic, metallo, serine, and threonine. It was developed at Shanghai Jiaotong University, China. ProtIdent uses a machine learning algorithm called OET-KNN trained on domain information from the CDD database along with SMART, Pfam, COG, and KOG to find functional domain uses matches between the query protein and known conserved domains. PsePSSM is also used to look at the evolutionary sequence patterns of the protein. The algorithm is set up as a "2-layer" predictor. The first layer is for identifying the query protein as protease or non-protease. If the protein is a protease, ProtIdent will go to the second layer to further identify it among the six types. The overall success rates in both cases by rigorous cross-validation tests were higher than 92%, but no confidence intervals or statistical values are provided in the results.⁶⁶

Comparisons of Function Predictions Tools

All the resources used are critical for predicting a protein's function. ProtIdent helps with this by identifying proteases through a machine learning model which has been trained on databases of conserved domains and patterns in evolution. That said, it is still quite limited because it is only useful for detecting proteases. Still it is a useful tool for comparing results from other sources. DeepGOWeb makes its predictions through neural networks and sequence similarities. Results also include confidence scores which can be changed to eliminate low confidence predictions. InterPro combines different databases to find similar proteins and their families, allowing for accurate function predictions. While ProtIdent is good at finding proteases and has a 92% success rate, it does not have any statistical results. DeepGOWeb provides more detailed predictions and a wider range of ability compared to ProtIdent. However it may occasionally give uncertain predictions if the protein sequence submitted is novel, similar to the one in this project. InterPro is very detailed and the best option of the three because it combines results from various databases and reduces false positives by comparing them with UniProtKB. Regardless, all three tools are great resources to use for protein function prediction.

Results

1. Sequence analysis

a. Functional/structural domains and motifs

Experiment	Outcomes	Amino Acid Position	Statistical Significance (e-value)	Appendix
CDD	Families: - Smac_DIABLO - Lipase - PLAT_PL - Tryp_SPc Domains: - Smac_DIABLO - Lipase - PLAT_PL - Tryp_SPc	- 39 - 269 - 271 - 605 - 608 - 718 - 719 - 954	- .97e-129 - 0e+00 - 1.03e-94 - 1.03e-94	A1
InterPro	Families: - Smac_DIABLO - Lipase Domains: - Smac_DIABLO - Lipase - PLAT - Tryp_SPc	- 40 - 269 - 271 - 605 - 608 - 721 - 719 - 954	N/A (Not provided)	A2
SMART	Domains: - Pfam: Smac_DIABLO - Pfam: Lipase - LH2 - Tryp_SPc	- 39 - 272 - 270 - 605 - 608 - 718 - 718 - 951	- 6.6e-102 - 3e-159 - 5.34e-23 - 3.14e-90	A3.1 , A3.2 , A3.2
HMMER	Domains: - Smac_DIABLO - Lipase - PLAT - Trypsin	- 39 - 275 - 270 - 605 - 610 - 716 - 719 - 951	- 3.3e-112 - 1.9e-162 - 1.9e-19 - 1.7e-67	A4

Final Consensus: On the basis of all these experiments, all four domains has been consistently predicted: Smac_DIABLO, Lipase, PLAT. and Tryp_SPc across common amino acid positions. HMMER and CDD has provided the most statistically significant results, with extremely low e-values, which indicates high confidence in domain assignments. Based on the e-values provided by each database and the overlap of the amino acid positions, the following locations appear most reliable: Smac_DIABLO (39- 271), Lipase (270-605), PLAT (608-718), and Tryp_SPc (719-951). The slight variation in domain naming, such as PLAT vs PLAT_PL vs LH2 or Tryp_SPc vs Trypsin, reflects different database-specific annotations of similar structural motifs. Therefore, these positions are most likely to represent the true conserved domains in the protein.

b. Sequence Motifs

Experiment	Outcomes	Statistical Significance	Appendix																																																						
NsitePred	Number of predicted binding residues: ATP: 24, ADP: 12, AMP: 15, GTP: 20, GDP: 10 ATP binding sites: <table><tr><td>80</td><td>F</td></tr><tr><td>81</td><td>G</td></tr><tr><td>332</td><td>S</td></tr><tr><td>345</td><td>H</td></tr><tr><td>347</td><td>F</td></tr><tr><td>424</td><td>G</td></tr><tr><td>425</td><td>S</td></tr><tr><td>444</td><td>G</td></tr><tr><td>473</td><td>H</td></tr><tr><td>484</td><td>G</td></tr><tr><td>486</td><td>G</td></tr><tr><td>493</td><td>H</td></tr><tr><td>497</td><td>F</td></tr><tr><td>665</td><td>L</td></tr><tr><td>748</td><td>C</td></tr><tr><td>810</td><td>D</td></tr><tr><td>816</td><td>L</td></tr><tr><td>830</td><td>C</td></tr></table>	80	F	81	G	332	S	345	H	347	F	424	G	425	S	444	G	473	H	484	G	486	G	493	H	497	F	665	L	748	C	810	D	816	L	830	C	Probability: <table><tr><td>0.098</td></tr><tr><td>0.183</td></tr><tr><td>0.105</td></tr><tr><td>0.107</td></tr><tr><td>0.198</td></tr><tr><td>0.405</td></tr><tr><td>0.206</td></tr><tr><td>0.281</td></tr><tr><td>0.277</td></tr><tr><td>0.098</td></tr><tr><td>0.169</td></tr><tr><td>0.105</td></tr><tr><td>0.098</td></tr><tr><td>0.147</td></tr><tr><td>0.103</td></tr><tr><td>0.253</td></tr><tr><td>0.199</td></tr><tr><td>0.190</td></tr></table>	0.098	0.183	0.105	0.107	0.198	0.405	0.206	0.281	0.277	0.098	0.169	0.105	0.098	0.147	0.103	0.253	0.199	0.190	A5
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Swiss-Model	Ligands: 1x CA 1 x Calcium Ion CA. 4 residues with 4Å: - Chain A: E.457, R.460, D.462, D.465	The binding site identified by	A6																																																																																										

	5 PLIP interactions: 5 interactions with chain A - Metal complexes: A:E.457, A:R.460, A:D.462, A:D.465, A:D.465 1NB (binding site not conserved) ACT, MPD (binding site not conserved)	Swiss-Model is supported by five PLIP-validated interactions within 4 Å of the calcium ion, indicating a statistically meaningful and spatially consistent binding site. More information including the bar chart is in the appendix.	
COACH	<u>COACH Results</u> Ligand: CA (PDB hit: 1w52X) Consensus binding residues: 457, 460, 461, 462, 465 Ligand: MUP (PDB hit: 1lpbB) Consensus binding residues: 346,347,421,422,423,448,450,485,533 <u>TM-SITE Results</u> Ligand: C8E Consensus binding residues: 345,346,347,348,349,355,381,421,422,522,526,529,534 Ligand: MUP, PLC Consensus binding residues: 346,347,421,422,423,448,450,485,533 <u>S-SITE Results</u> Ligand: CA Consensus binding sites: 457,458,460,461,462,465 Ligand: PLC, MUP Consensus binding sites:	C- Score: 0.27 C- Score: 0.09 C- Score: 0.34 C- Score: 0.16 C- Score: 0.20	A7

	346,347,348,384,421,422,423,448,450,483,485,522,525,526,533,534 Ligand: IOD, 1NB Consensus binding sites: 746,747,763,900,901,902,903,904,905,924,925 <u>COFACTOR Results</u> Ligand: CA Consensus binding sites: 457, 460, 462, 465 Ligand: BMA Consensus binding sites: 433, 434 Ligand: MUP Consensus binding sites: 346, 347, 421, 422, 423, 448, 450, 485, 533 Ligand: C8E Consensus binding sites: 345, 346, 347, 348, 349, 355, 381, 421, 422, 534 Ligand: DDQ Consensus binding sites: 348, 349, 384, 485, 533	C- Score: 0.12 C- Score: 0.11 C- Score: 0.47 C- Score: 0.34 C- Score: 0.33 C- Score: 0.15 C- Score: 0.14 C-score is the confidence score of predicted binding	
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		site. C-score values range in between [0-1]; where a higher score indicates a more reliable prediction. A score of 0.25 is considered moderately confident.	
Final Consensus: The predicted binding sites falls into two major categories: nucleotide binding sites and ligand-binding sites. NsitePred identified a range of nucleotide-binding sites. Among these, several high confidence sites (probability > 0.35) are ATP: 424(G), AMP: 345(H), 422(S), GTP: 490(T), GDP: 425(S). 422 (S) had the highest binding probability for AMP at 0.515, and 425 (S) had 0.902 for GDP, indicating their strong relevance in phosphate group interactions. In contrast, ligand-binding sites, identified by COACH and SWISS-MODEL, showed different pattern. The calcium binding pocked is particularly well-conserved, with residues E457, R460, D462, and D465 consistently predicted by both Swiss-Model and COACH. These residues do not overlap with the nucleotide-binding clutter and seem to form a distinct metal coordination site. COACH also predicted several ligand-binding sites for small organic molecules such as MUP, PLC, and CBE. In conclusion, the calcium-binding site (457–465) is structurally isolated and highly conserved, supporting strong ion coordination.			

c. Secondary Structures

Experiment	Outcomes	Statistical Significance	Appendix
PSIPred	MLRAALSTARRGPRLSRLLSAAATSAV PAPMAALKSWLSRSVTSFFRYRQCLCV PVVANFKKRCFSELRPWHKTVTIGFGV TLCAVPPIAQKSEPHSLSS EALMRRVSL VTDSTSTFLSQTTYALIEAITEYTKAVYT LTSLYRQYTSLLGKMNSEEEDEVWQVI IGARAEMTSKHQEYLKLETTWMTAVG LSEMAAEAAYQTGADQASITARNHIQL VKLQVEEVHQLSRKAETKLAEAQIEEL RQKTQEEGEERAESEQEAYLREDNEVC YERLGCFSDDSPWAGIVERPLKILPWSP EKVNT RFLLYTNENPDNFQEIVADPSTI	The confidence of each amino acid prediction is shown as a bar chart in the appendix.	A8

	QSSNFNTGRKTRFIIHGFIDKGEESWLS TMCQNMFKVESVNCICVDWKS GSRTA YSQASQNV RIVGA EVAYLVGV LQSSFD YSPSNVHIIGHSLGSHAAGEAGRRTNG AVGRITGLDPAEPCFQGTPELVR LDPSDAQFVDVIHTDIAPFIPNLGFGMS QTAGHLDFFPNGGKEMPGCQKNVLSQI VDIDGIWQGTRDFAACNHLRSYKY YT DSILNPDGFAGFSCASYSDFTANKCFPC SSEGCPQMGHYADRFPGRTKGVGQLF YLN TGDA SNFARWRYRVDVTLSGKKV TGHVLVSLFGNKGNSRQYEIFQGT LKP DNTYSNEFDS DVEVG DLEKVKFIWYN NVINLTLPKVGASKITVERNDGSVFNFC SEETVR EDVLT LTACVVGGEDARPHS WPWQISLQYLN D TWRHTCGGT LIAS NFVLTAAHCAISNTRTYRVAVGKNNLEV EDEEGSLFVGVDTHVHKRWNALLLR NDIALIKLAEHVLSDTIQVACLPEKDS LLPKDYP CYVTGW GRLWTNGPIADKL QQGLQPVDHATCSRIDWWGFRVKKT MVCAAGDGVISACNGDSGGPLNCQLE NGSWEVFGIVSFGSRRGCNTRKKPVVY TRVSAYIDWINEK  : Beta Strand  : Alpha Helix : Coil		
JPred4	MLRAALSTARRGPRLSRLLSAAATSAV PAPMAALKSWLSRSVTSFFRYRQCLC VPVVANFKKRCFSELIRPWHKTVTIGF GVTLCAPVIAQKSEPHSLSSEALMRRA VSLVTDSTSTFLSQTTYALIEAI --HHHHHHH----HHHHHHH-----HHHH HHHHHHHHHHHHHHHHHHHHHHH---EHHHH HHHHHHHHH---HHHHHHH---EEEE----- -----HHHHHHHHHHHHHHHH---HHHHHHHH	The JNETCONF confidence of each amino acid prediction is shown as a bar chart in the appendix.	A9.1 , A9.2

	HHHHHHHHHHHHHHHHHHHHHHHTEYTKA VYTLTSLYRQYTSLLGKMNSEEEDEV WQVIIGARAEMTSKHQEYLKLETTWM TAVGLSEMAAEAAAYQTGADQASITARN HIQLVKLQVEEVHQLSRKAETKLAEAQI EELRQKTQEEGEERA ESEQEAYLREDNEVCYERLGCFSDDS PWAGIVERPLKILPWSPEKVNTFRLLY TNENPDNFQEI HHHHHHHHH-----HHHHHHHHHHHHHH HHHHHHHHHHHHHHHHHHHHHHHHHHHHHH HHHHHHHHHH---HHHHHHHHHHHHHHHHHH HHHHHHHHHHHHHHHHHHHHHHHHHHHH-HHH HHH-----VADPS TIQSSNFNTGRKTRFIIHGFDKGEESW LSTMCQNMFKVESVNCICVDWKSGS RTAYSQASQNVRIVGAEVAYLVGVLQS SFDYSPSNVHIIGHSLGSHAAGEAGR TNGAVGRITGLDPAEPCFQGTPELVRL DPSDAQFVDVIHTDIAPF -----EEEEEE-----EE----- -----EEEEEE-----HHHHHHHHHH HH---EEEEEE-----HHHHHHHHHHHH HHHHHHHHHHHHHHHHHH-----EEEE---HHH HHHHHHHHHH-----EEE----- EEEEEE----- IPNLGFGMSQTAGHLDFFPNGGKEMP GCQKNVLSQIVDIDGIWQGTRDFAAC NHLRSYKYYTDSILNPDGFAGFSCASY SDFTANKCFPCSSEGCPQMGMHYADRF PGRTKGVGQLFYLN TGDA SNFARWR YRVDVTLSGKKVTGHVLVSLFGNKGN SRQYEIFQGTLKPDNTYSNEFDSDVE VGDLEK -----EE-----EEEEEE----- ---E---EEEEEE-----EEEEHH----- ----- -----HHHHHH-----		
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	-----VKFIWY NNVINLTLPKVGASKITVERNDGSVFN FCSEETVREDVLLTLTACVVGGEDARP HWPWQISLQYLKNDTWRHTCGGTLI ASNFLTAAHCISNTRTYRVAVGKNNL EVEDEEGSLFVGVDTI -----EEEE-----EEEE EEEE-----EEEE-----EEEE----- --EEEEEEEEHVHNRWNALLLRNDIAL IKLAEHVELSDTIQVACLPEKDSLLPKD YPCYVTGWGRLWTNGPIADKLQQGL QPVDHATCSRIDWWGFRVKKTMC AGGDGVISACNGDSGGPLNCQLENG SWEVFGIVSFGSRRGCNTRKKPVVYT RVSAYIDWINEK E-----HHHHH-----EEEE----- -----EEEEEEEE-----EEE----- -----EEEE-----EEEE EEEE-----EEEE-----H----- E: Beta Strand H: Alpha Helix --: Coil/Unstructured		
PREDATOR	MLRAALSTARRGPRLSRLLSAAATSAVPAPM AALKSWLSRSVTSFFRYRQCLCPVAVNFKK RCFSELIR ccchhhhhhhccchhhhhhhccccccch hhhhhhhccccccccccccccccchhh hhhhhhcc PWHKTVTIGFVTLCAVPIAQKSEPHSLSSE ALMRAVSLVTDSTSTFLSQTTYALIEAITE YTKAVYTL ccccccccccccccccccccccccchhh hhhhhhhhhhccccccccchhhhhhhhhhh hhhhheeee TSLYRQYTSLLGKMNSEEEDEVWQVIIGARA EMTSKHQEYKLETTWMTAVGLSEMAAEAY QTGADQAS eeccccceccccccccchhhhhhhhhhhhh hhhhhhhhhhhhhhhhhhhhhhhhhhhhhh	The Reliability Graph ($0 < r < 1$) for each amino acid residue is shown in the appendix.	A10

hcccchhh ITARNHIQLVKLQVEEVHQLSRKAETKLAEA QIEELRQKTQEEGEERAESEQEAYLREDNEV CYERLGCF hhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhh hhhhhhhhhhhhhhhhhhhhhhhhhhhhhhccee ecccccee SDDSPWAGIVERPLKILPWSPEKVNTRFLLY TNENPDNFQEIVADPSTIQSSNFNTGRKTRF IIHGFIDK ecccccccccccccccccccccccccceeee ecceeeeeeeccc hhhhhh cccccceeee eecccccc GEESWLSTMCQNMFKVESVNCICVDWKSGSR TAYSQASQNVRIVGAEVAYLVGVLQSSFDYS PSNVHIIG ccc hhhhhhhh cccccceeeeecccccc cc hhhhhhhhhhhhhhhhhhhhhhhhhhhhhccc ccceeeee HSLGSHAAGEAGRRTNGAVGRITGLDPAEPC FQGTPELVRLDPSDAQFVDVIHTDIAPFIPN LGFGMSQT ec hhhhhhhhhhhh cccccceeeeecccccc ccccccccccccccccceeecccccccccc cccccccc AGHLDFFPNGGKEMPGCQKNVLSQIVDIDGI WQGTRDFAACNHLRSYKYTDSILNPDGFAG FSCASYSD ccc eeeee ccccccccccccccccccc hhhh hh ccccccc hhhhhhhhhhhhhh cccccccc cccc hhh FTANKCFPCSSEGCPQMGHYADRFPGRTKGV GQLFYLNTGDASN FARWRYRVDVTL SGKKVT GHVLVSLF hh ccccccccccccccccccccccccccc ceeeeeccccccccceeeeeeeeeeeccc eeeeeeee GNKGNSRQYEIFQGTKPDNTYSNEFDS DVE VGDLEKVKFIWYNNVINLTLPKVGASKITVE RNDGSVFN ecccccccceeeeeeeccccceeeeeeeccc ccceeeeeeeeeccccccccceeeeeeee	
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	ecccccee FCSEETVREDVLLTLTACVVGGEDARPHSWP WQISLQYLKNDTWRHTCGGTLIASNFVLTA HCISNTRT ecccccccccccccccccchcccccccccccc cccccccccccccccccccccccccccccccc ccccccce YRVAVGKNNLEVEDEEGSLFVGVDTIHVHKK WNALLLRNDIALIKLAHVELSDTIQVACLP EKDSLPLK eeeecccccccccccccccccccccccccccc hhhhhhhchhhhhhhhhhhcccccccccccc cccccccc DYPCYVTGWGRLWTNGPIADKLQQGLQPVVD HATCSRIDWWGFRVKKTMVCAGGDGVISACN GDSGGPLN cccccccccccccccccccccccccccccccc cccccccccccccccccccccccccccccccc cccccccc CQLENGSWEVFGIVSFGSRRGCNTRKKPVVY TRVSAYIDWINEK cccccccccccccccccccccccccccccccc ecccccccccccc e: Extended Strand, 207 is 21.70% c: Random Coil, 481 is 50.21% h: Alpha Helix, 266 is 27.88%		
SCRATCH	<u>Predicted Secondary Structure (3 Class):</u> CCHHHHHHCCCCCHHHHHHHHHHHCCCCCHHH HHHHHHHHHHHHHHCHCCCCCHHHCCCCC CCCCCCCCCHCCEEECECHHCCCECCCCC CCCCCHHHHHHHHHHHHHHHHHHHHHHHHH HHHHHHHHHHHHHHHHHHHHHHHHHHHHCCCC HHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHH HHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHH HHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHH ECCCCCCCCCCCCCCCCCCCCCHHHCCCCCEE EECCCCCCCCCCCCCCCCCCCCCHHHHHCCCCC EECCCCCCCCCCCCCCCCCCCCCHHHHHCCCCC EECCCCCCCCCCCCCCCCCCCCCHHHHHCCCCC	Based on prior validations, SCRATCH achieves ~84% accuracy in class 3 prediction and ~72% in class 8 prediction.	A11

	<pre> EEEECCCHHCCCHHHHHHHHHHHHHHHHHHHHH HHHHHCCCCCHHHEEEEEEECHHHHHHHHHHHHH CCCCCCEEEEECECCCCCCCCCCCCCCCCCHHH CCCEEEECCECCCCCHHHHCCCECCCCCCCCEEE ECCCCCCCCCCCCCHHHHHCCCHCCCHCCCHH HHHHHHHHHHHHHHHHHHHHCCCCCECEECCH HHHHCCCCCCCCCCCCCECCCCCHHCCCCC CCCEEEECCECCCCCCCCCEEEEEEEEEEEEC EEEEEEEEEEECCECCCEEEEEEEEEECCE EEEEEEECCECCCCCEEEEEEEEEECCECC CEEEEEEEEEECCECCCCCEEEEEECCECC EEECCECECECCCCCCCCCEEEEEEEEEECCE EEEEEEEEEEECCEEECHHHCCCCCEEEEE CCCECCCCCCCCCEEECEEEEEEECCCCCCC CCCCCEEEEECCCCCCCCCCCCCECCCCC CCCCCEEEEECECCCCCECECEEEEEEC EECHHHHCCCCCHHHCCCCCEEECCCCCE CCCCCCCCEEEEECCEEEEEEEEEEEEC ECCECECEEEEEHHHCHHHHHH </pre> Predicted Secondary Structure (8 Class): <pre> CHHHHHHHHHCHHHHHHHHHHHHHCCCCCHH HHHHHHHHHHHHHHHHCHTTCEEEEECCCCS CCCCSCCCHHHHHHHHHHHHHHCCECCCC CCCCCHHHHHHHHHHHHHHHHHHHHHHHHH HHHHHHHHHHHHHHHHHHHHHHHHHHHTTSS HHHHHHHHHHHHHHHHHHHHHHHHHHHHHH HHHHHHHHHHHHHHHHHHHTTCHHHHHHHHH HHHHHHHHHHHHHHHHHHHHHHHHHHHHHH HHHHHHHHHHHHHHHHHHHHHHHCCEEEETT EECCTTTSSBTTBCCCCCCCCCHHHHCCEEE EETTEEEEEEEECSCHHHHHHSSCCTTSEE EEECCCCCTTCTTHHHHHHHHHHHHCCEEE EEECHHHHSSCHHHHHHHHHHHHHHHHHHH HHHHHHCCCCGGGEEEEETHHHHHHHHHHH TTTCSSEEEEESCBCTTTTTTSCTTTSCGGG SSEEEEECSCHHHHCCBCSSCCSSEE EETTSSSCTTCCCCCCCCSSCCHHHHHHTSS CCCHHHHHHHHHHHHHHHCTTTTBCEECCH HHHTTTCSCSCCTTCCCBSSTTGGGSTTTT SSSEEEEECCCCSSSSCCCEEEEEEEEEEEEC EEEEEEEEEEETTTECCCEEEEEEEECSSCE EEEEEEESSCCCSEEEEEEEEEESSCTTSC CEEEEEEEEEETTSCEEEEECCCCBCTTCCE EEEECCBSCEECCTTSSTTEEEEEEEETTEE EEEEEEEEEEETTTEEECGGGCCTTSCEEEEE SCSBTTCTTCTTCEEECEEEEEECTTCCTTT CTTCCEEEEESSCCCCSSCCCBCCCTTCC </pre>		
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	CCTTCCEEEEEESCBSSSTTCCBCSBCEEEEC EECHHHHTSTTTTGGGCCTTEEEECSSSSB CCTTCTTCEEEECTTSCEEEEEEEEEEBTT BSSBTTBCEEEEEEGGSHHHHHH H: Alpha Helix E: Beta Sheet C: Coil/Unstructured		
Final Consensus: The secondary structure predicted by PsiPred, PREDATOR, JPred4, and SCRATCH are roughly similar. The first 270 amino acids residues have strong alpha-helical region, aligning with the Smac_DIABLO domain. This prediction is consistent throughout all the experiments. The Lipase domain residues (~217 - 605) displays a mixed pattern of some alpha helices and mostly beta strands. Further down in the C terminus, the Try_Spc (~719 - 954) shows an even more pronounced beta stands content. The consistency across methods enhances confidence in these structural assignments and supports their functional implications in catalysis, binding, and regulation.			

d. Hydrophilic/Hydrophobic Regions

Experiment	Outcomes	Locations	Statistical Significance	Appendix
ePESTfind	A total of 12 PEST motifs were located in the amino acid sequence All PEST motifs were found in positions 275 and above 'Valid' PEST motifs below the threshold score (5.0) were tabled as 'poor', while PEST scores above the threshold were considered as potential PEST motifs	276 -292 307 -338 392 -417 441 -460 460 -473 473 -493 538 -565 565 -579 647 -667 819 -834 840 -851 887 -929	-13.02 -5.08 -19.10 -6.99 -15.47 -17.04 -10.78 -15.47 6.01 -12.03 -24.30 -13.18	A12
Expasy	Sustained stretch peaking above +1 (Hydrophilic)	~10–40	Not available	A13

	Noticeable peak beyond -2 (Hydrophobic) Peak decreasing from -1 to -1.5 (Hydrophobic) Noticeable peak at -3 (Hydrophobic) Strong sharp peak almost reaching +2 (Hydrophilic) Sustained values at or below -1 (Hydrophobic) High peak almost reaching +3 (Hydrophilic) Peaks at or below -1 (Hydrophobic) Sustained dip near -2 (Hydrophobic)	~120–150 ~180–240 ~250–310 ~370–410 ~530–580 ~690–730 ~770–800 ~820–870		
SCRATCH Protein Predictor-A CPro	Regions of 6 or more Hydrophilic Regions (e) in a row 8 8 6 18 6 6	21–28 81–88 291–296 301–318 451–456 721–726	Not provided	A14

	Regions of 6 or more Hydrophobic Regions (-) in a row			
	7	29–35		
	8	139–146		
	8	149–156		
	8	163–170		
	8	211–218		
	8	234–241		
	8	259–266		
	8	345–352		
	8	371–378		
	8	495–502		
	8	545–552		
Final Consensus: From the three different tools used to predict hydrophobicity within the novel protein, the results find a mix of hydrophobic and hydrophilic regions, but with a little more hydrophobicity overall. Expasy supports these findings, detecting several sustained and strong hydrophilic peaks reaching +2 and +3 in some regions, and strong and sustaining hydrophobic dips down to -2 and -3 in others. ePESTfind, however, identified PEST motifs located in the C-terminal half of the protein. However, SCRATCH Protein Predictor – ACCPro combines machine learning methods and evolutionary libraries to detect stretches of accessible hydrophobic and hydrophilic residues, allowing its findings are more supported.				

e. Trans-membrane domains

Experiment	Outcomes	Amino Acid Position	Statistical Significance	Appendix
TOPCONS	6 database comparisons state that it has no Trans-membrane domains	Not provided	1.0	A15
DeepTMHMM	States that it has no Trans-membrane domains	1-490	1.0	A16
TMAP	States that it has no Trans-membrane domains	1-950	Not provided	A17
Final Consensus: Based on the unanimous decision from all three databases, it is predicted that				

the novel protein lacks any trans-membrane domains. Statistical significance scores of 1.0 reported from both DeepTMHMM and TOPCONS greatly increases the reliability of the prediction, since TOPCONS utilizes six different databases, while DeepTMHMM utilizes deep learning methods.

f. Subcellular Localization / protein-sorting signals

Experiment	Outcomes	Amino Acid Position	Statistical Significance	Appendix
BUSCA	Extracellular	Not provided	1.0	A18
WoLF PSORT	Mitochondria	Not provided	15	A19
ESLPred	Extracellular	Not provided	Reliability Index= 4 Expected Accuracy=~ 97%	A20
DeepLoc2.1	Extracellular	1-30	0.8469	A21
IPSortProdiction	Mitochondrial	1-30 61-69	0.083	A22
TargetP 2.0	Mitochondrial Peptide Transferase	17-20	0.9573	A23
CELLO	Extracellular	Not provided	0.932	A24
HHPred	Identified as being closely related to “PANCREATIC LIPASE RELATED PROTEIN 1; HYDROLASE, LIPID DEGRADATION, PANCREATIC LIPASE” from the species <i>Canis</i>	270-718	Probability: 100% E-Value:7.3e-7 Score:687.91 SS: 50.1	B1

	<i>lupus</i>			
Final Consensus: Based on the combined predictions from the seven tools, it may be possible that the novel protein has two subcellular localizations in both the extracellular region and the mitochondria. BUSCA, ESLPred, DeepLoc2.1, and CELLO, all support the prediction that the protein has an extracellular localization. DeepLoc2.1 reported the presence of a signal peptide within amino acids in positions 1–30, which is the region, specifically the N-terminus, characteristic of proteins that are secreted. DeepLoc2.1, also assigned a relatively high confidence score of 0.8469. CELLO supports this score with its own, at 0.932. But, while there is less statistical support from the tools predicting the protein having a mitochondrial subcellular localization, evidence from WoLF PSORT, IPSortPrediction, and TargetP 2.0 suggests that mitochondrial localization may be a possibility. TargetP 2.0 assigns a high confidence score of 0.9573 to the prediction that there is mitochondrial peptide transferase activity within amino acids 17–20, which is an area that overlaps with the predicted signal peptide region for extracellular targeting. This overlap indicates the possibility that the protein may be initially directed toward the mitochondria then move to the extracellular region, or may function in both environments depending on cellular context or under specific conditions. HHPred allows for accurate predictions				

g. Function Predictions

Experiment	Outcomes	Amino Acid Position	Statistical Significance	Appendix
ProtIdent	Protein is a protease and belongs to Serine type	1-954	Not provided	A25
DeepGOWeb	Predicted to have hydrolase an catalytic activity	1-954	Not provided	A26
InterPro	Predicted to be involved in apoptotic process, proteolysis, and lipid catabolic process	1-954	Not provided	A27
Final Consensus: Based on the results from the three tools, the protein is predicted to be a serine hydrolase that regulates apoptosis. ProtIdent predicted that the protein is a serine-type protease, while DeepGOWeb indicated that the protein has hydrolase and general catalytic activity. Finally, InterPro predicted the protein most likely participates in apoptosis, proteolysis, and lipid catabolism. Based on these results, the protein's function suggests that it plays a critical role in controlled cell death and the breakdown of proteins and lipids.				

Domains Table

Database	Name of Functional Domain/ Motif	Amino Acid Position	Known Standard Sequence Motif Arrangement	Unique Identifying Features	Examples of Domains/Motifs in other proteins
CDD	SMAC DIABLO	39-269	None present	Interacts with Inhibitor of Apoptosis Proteins (IAPs)	1. Omi/HtrA2 is a mitochondrial serine protease that does mitochondrial homeostasis and apoptosis.⁷⁰ 2. ARTS is a protein that suppresses tumors in Acute Lymphoblastic Leukemia (ALL).⁷¹
	Lipase	271-605	Catalytic Triad: SDH Positions 422, 446, 533 Nucleophilic Elbow: GHSLG Positions: 420-425 Consensus:G-X-S-XG ⁶⁸	Active Site Positions 507,508 530,531 CQ, AC	1. Elastase is a serine protease with a catalytic triad that is made by the pancreas.⁷² 2. Chymotrypsin is a digestive enzyme that has a nucleophilic elbow. It's synthesized in the pancreas and breaks down polypeptides.⁷³
	PLAT/LH2	708-718	None present	Interacts with procolipase ⁷⁰	1. ALOX5 is a protein that converts arachidonic acid into leukotrienes. Leukotrienes are lipid mediators involved in

					inflammation and allergic reactions.⁷⁴ 2. Triglyceride lipase is a pancreatic enzyme that is found in the early larvae stages of the Japanese Eel.⁷⁵
	Trypsin-like serine protease	719-954	None present	Cleavage Site Position 719 V Active Site Positions 899,925,927 HDS Substrate Binding Site Positions 899, 925,927 SSG	1. Prothrombin is a protein found in domestic cattle that plays a role in blood clotting.⁷⁶ 2. Acrosin is a protein found in the acrosome of spermatozoa that acts as a hydrolytic enzyme that digests and penetrates an egg's zona pellucida during fertilization.⁷⁷

3D Structures of Known Domains

Smac_DIABLO

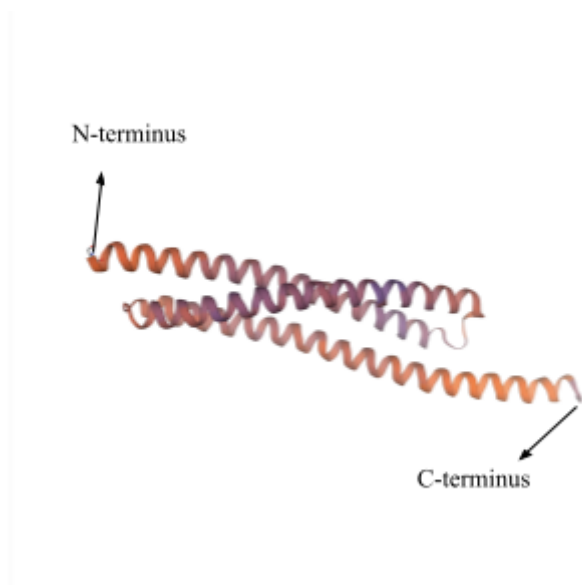


Figure 1.1. 3D structure of Smac_DIABLO ([A28](#))

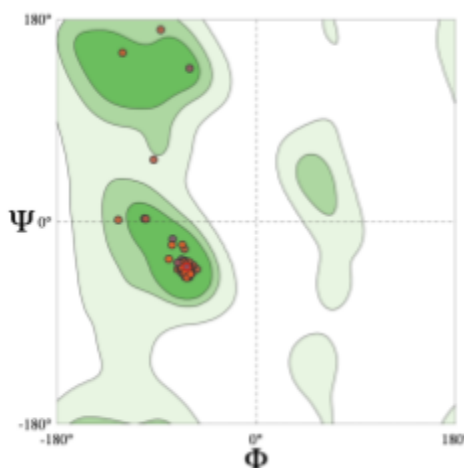


Figure 1.2. Ramachandran Plot of the Smac_DIABLO motif ([A28](#))

Figure 1.1. 3D structure of the Smac_DIABLO motif predicted using homology modeling through SWISS-MODEL (ExPASy platform). The model was generated based on sequence alignment to structurally characterized homologs available in the Protein Data Bank (PDB).

Figure 1.2. Ramachandran plot of the Smac_DIABLO domain. Most residues fall within the favored regions, indicating good stereochemical quality and supporting the reliability of the predicted 3D structure.

Lipase

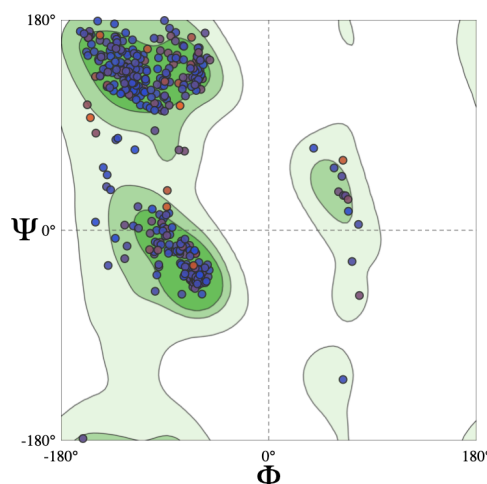
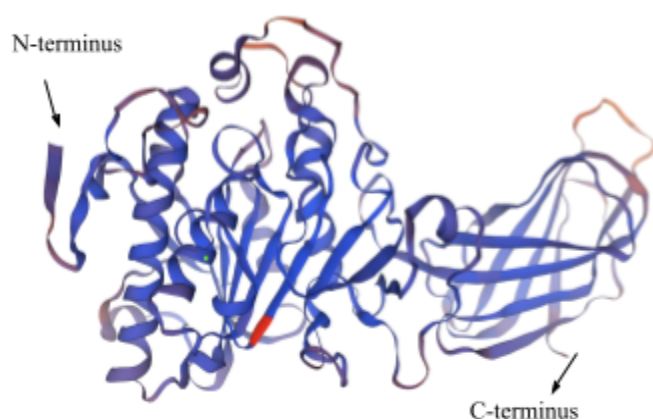


Figure 2.1. 3D structure of the LIPASE domain([A30](#)) **Figure 2.2.** Ramachandran Plot of the LIPASE domain ([A30](#))

Figure 2.1. 3D structure of the Lipase domain predicted using homology modeling through SWISS-MODEL (ExPASy platform). The model was generated based on sequence similarity to known lipase structures deposited in the Protein Data Bank (PDB). The structure displays a mixed composition of alpha helices and beta sheets, characteristic of the α/β -hydrolase fold typical in lipase enzymes.

Figure 2.2. Ramachandran plot of Lipase domain. Most residues fall within the favored regions, indicating good stereochemical quality and supporting the reliability of the predicted 3D structure.

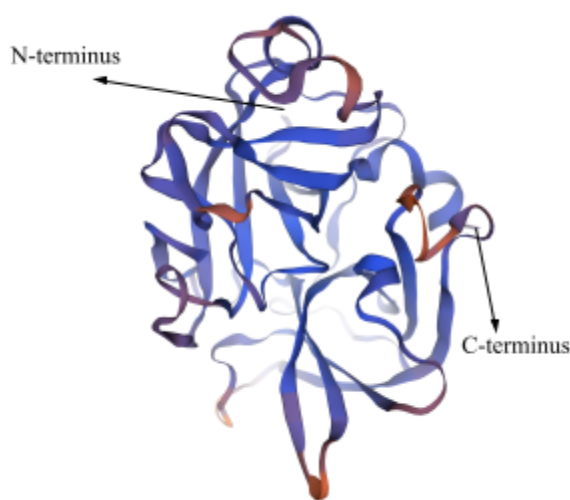
Tryp_SPc

Figure 3.1. 3D structure of the Tryp_SPc domain ([A31](#))

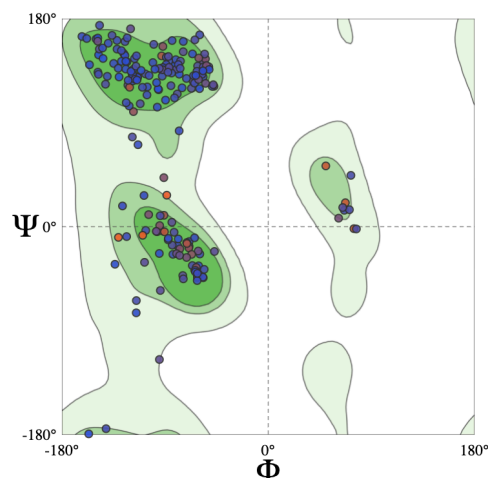
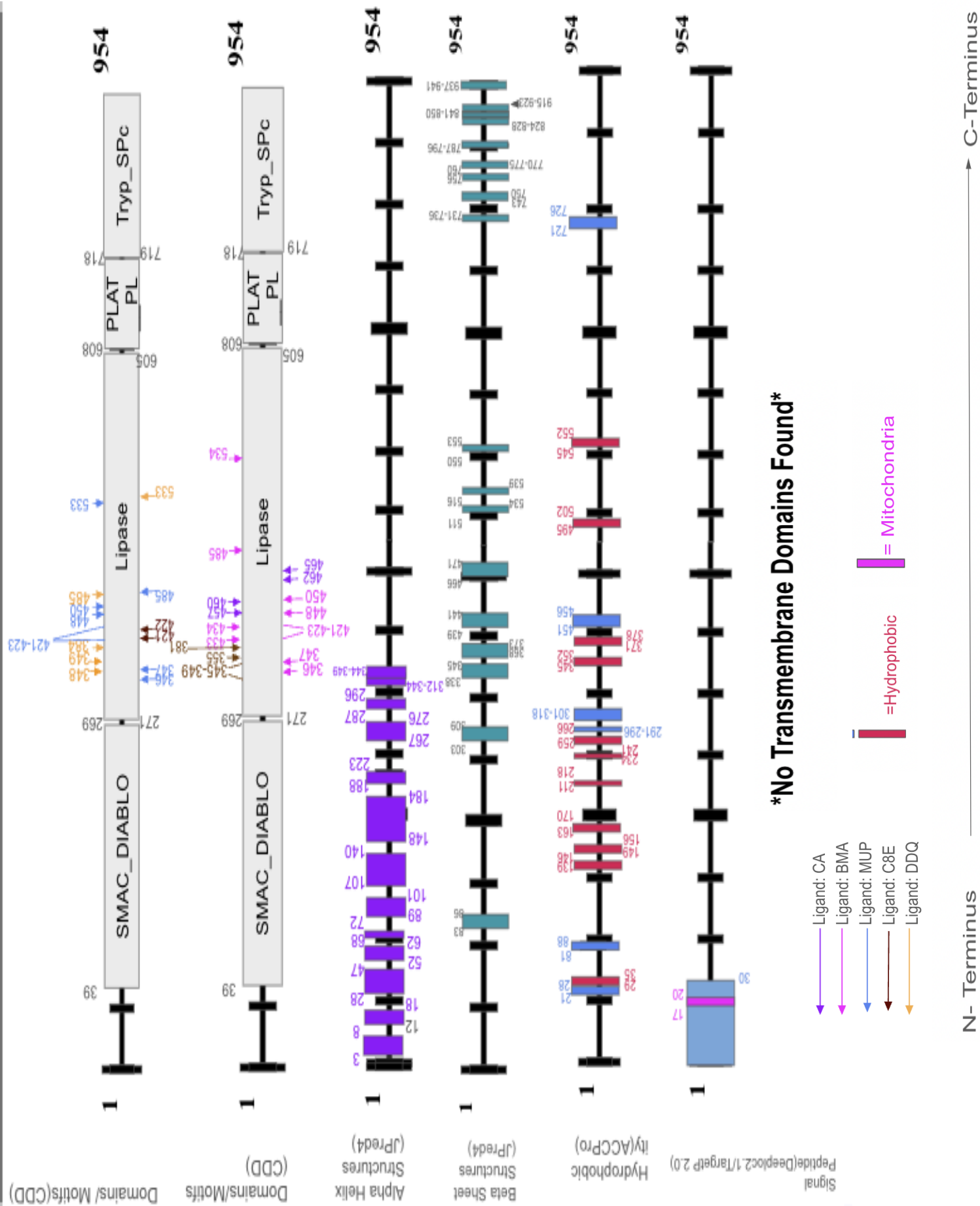


Figure 3.2. Ramachandran Plot of the the Tryp_SPc ([A31](#))

Figure 3.1. 3D structure of the Tryp_SPc domain generated using homology modeling via SWISS-MODEL (ExPASy platform). The predicted structure was built based on templates with high sequence similarity from the Protein Data Bank (PDB). The domain predominantly adopts a β -barrel-like fold, characteristic of serine proteases such as trypsin, supporting its catalytic role in protein degradation.

Figure 3.2. Ramachandran plot of the Tryp_SPc domain. Most residues fall within the favored regions, indicating good stereochemical quality and supporting the reliability of the predicted 3D structure

Composite Diagram



4. 3D Structure of Protein

i. I-TASSER Result

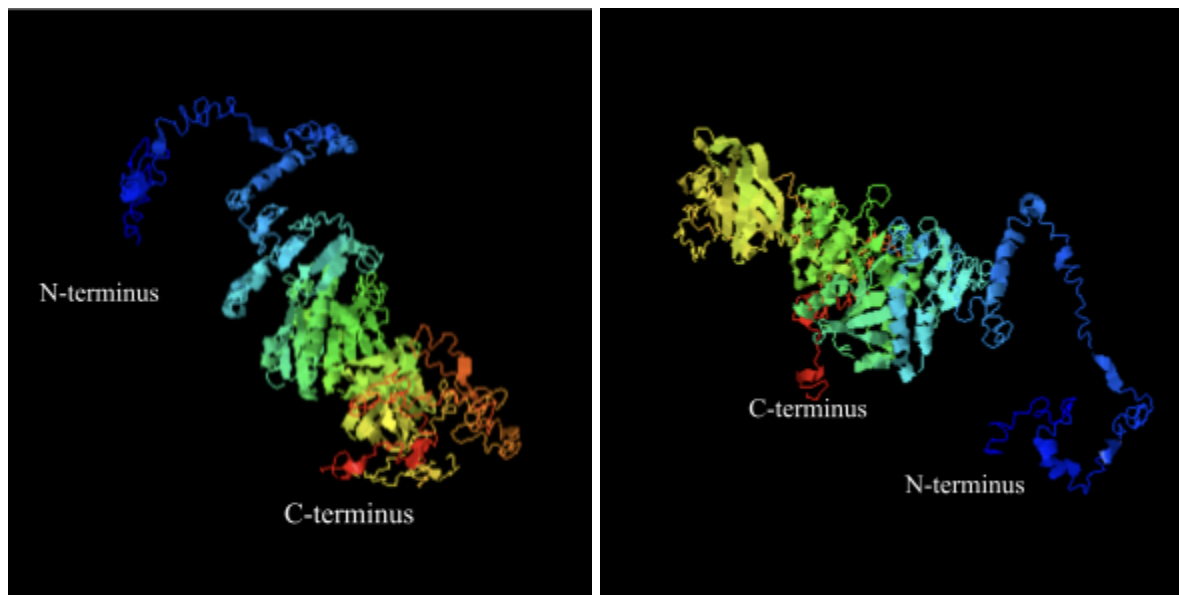


Figure 4.1 and Figure 4.2: 3D Structures of the Protein. Dark blue: Smac_DIABLO domain, Light blue and Green: LIPASE domain, Yellow and Orange: Tryp_SPc domain ([A34](#))

The 3D structure of the protein was predicted using I-TASSER (Iterative Threading ASSEmby Refinement), developed by the Zhang Lab. I-TASSER employs a hierarchical modeling strategy, starting with the identification of suitable structural templates from the Protein Data Bank (PDB) through a multiple-threading algorithm. Following template recognition, the server performs iterative fragment assembly simulations to build the full-length atomic model of the protein. Functional insights are then inferred by structurally aligning the predicted model with known proteins in the database.⁴² In the generated model, the N-terminus is shown in blue and the C-terminus in red to indicate directionality. Additionally, **Figure 4.2** displays the same protein structure as **Figure 4.1**, but rotated 180 degrees to offer an alternative perspective on the overall folding and spatial arrangement.

ii. AlphaFold Result

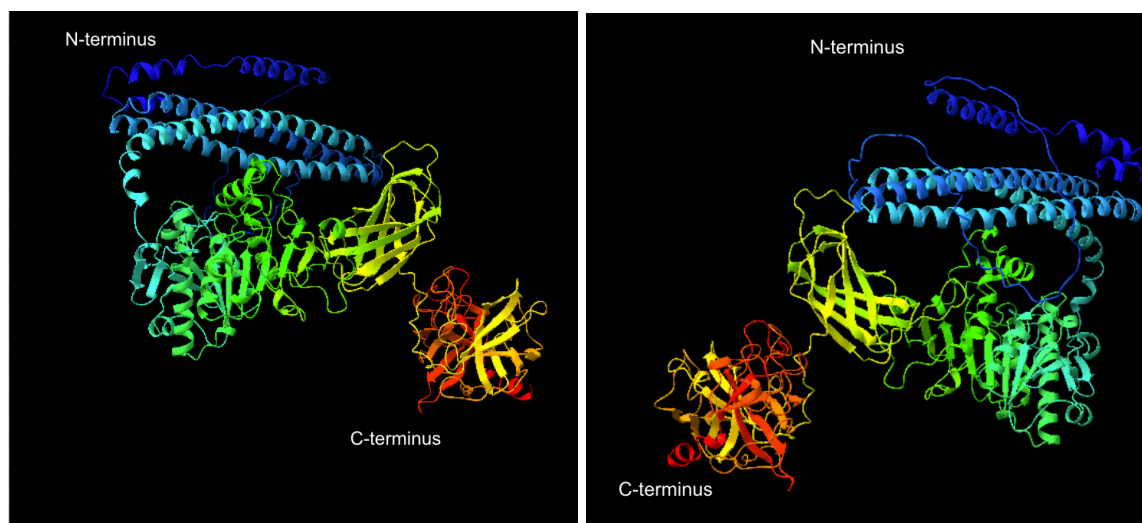


Figure 5.1 and Figure 5.2: 3D Structures of the Protein made by AlphaFold and Rendered in ChimeraX. Dark blue: Smac_DIABLO domain, Light blue and Green: LIPASE domain, yellow and Orange: Tryp_SPc domain ([A35](#))

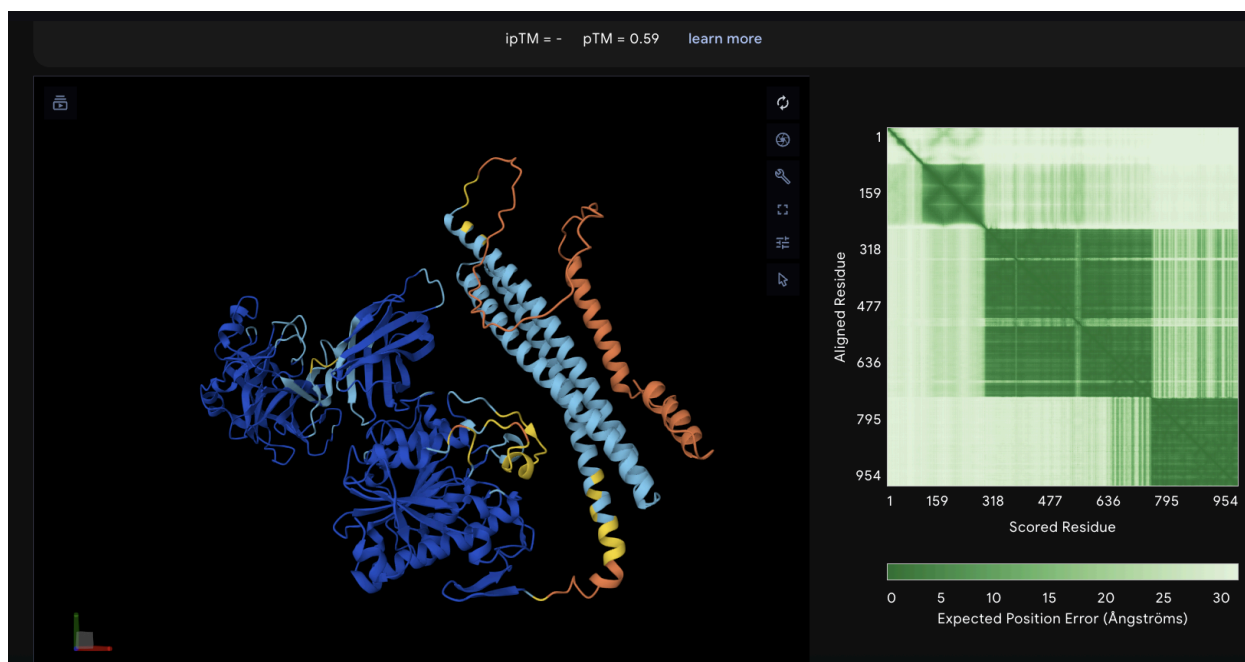


Figure 5.3: 3D Structures of Protein made by AlphaFold with Dark Blue Indicating Very high confidence, (pLDDT > 90) light blue meaning confident, (90 > pLDDT > 70) yellow meaning low confidence, (70 > pLDDT > 50) and orange meaning very low confidence (pLDDT < 50). pTM score is also provided ([A36](#)).

AlphaFold is an AI based system that predicts the 3D structure of proteins by comparing query amino acid sequences to known protein structures. **Figure 5.1** and **Figure 5.2** were generated using AlphaFold and were then rendered in ChimeraX using the rainbow tool, which colors the residues in the structure blue at the N-terminus to red at the C-terminus.⁶⁰ In Figure 5.3, the pTM and ipTM scores are provided. Alpha Fold uses statistical scores like pTM (predicted template modeling score) and ipTM (interface predicted, to measure prediction accuracy. A pTM score over 0.5 suggests the overall shape is likely accurate, while ipTM checks how well residues are positioned. Predicted Aligned Error (PAE) is also provided, and shows confidence levels regarding the position of each residue relative to one another in the predicted structure. Specifically, it measures the expected distance error, in Ångströms (Å), to see if the structure is aligned in a probable way.⁶¹

Rationale of using I-TASSER and Alpha Fold

I-TASSER predicts structure based on template-based modeling by threading known PDB structures, making it ideal when good templates exist.

AlphaFold, in contrast, uses deep learning and evolutionary information to predict structures even without close homologs, offering high accuracy for novel folds.

Using both methods allowed cross-validation. I-TASSER gave template-based confidence, while AlphaFold ensured deep-learning based refinement for potentially unknown regions.

Discussions and Conclusions

a. Correlation between previously well-established unique, identifying features

The protein interacts with a diverse set of small molecules and ligands, each contributes uniquely to its functional profile. Small molecules such as ATP, SMP, GTP, and ligands like calcium, MUP, and CBE, all contribute to the catalytic and regulatory functions. Calcium ions are known to play essential roles in enzyme activation, membrane binding, and structural stability, particularly in lipase enzymes. The protein also contains several well-conserved functional domains—Smac_DIABLO, Lipase, PLAT, and Trypsin. The Smac_DIABLO is known to interact with Inhibitor of Apoptosis proteins (IAPs), promoting apoptosis by neutralizing IAP function.⁸³ The presence of it also seen in similar mitochondrial proteins like Omi/HtrA2, which also regulates mitochondrial homeostasis and programmed death (refer to the Domains table). The lipase domain features conserved motifs including the catalytic triad SDH, and a nucleophilic elbow (G-H-S-L-G), both are essential for enzymatic activity. These are well documented in other lipases like Elastase and Chymotrypsin, which also share the catalytic triad or elbow for hydrolytic function (refer to the Domains table). The PLAT domain, on the other hand, is implicated in membrane association and is often found in lipid-related proteins like ALOX5. ALOX5 turns a fat molecule called arachidonic acid to leukotrienes, which are chemical messengers that causes inflammation and allergic reactions. It is likely that this domain helps the protein interact with the membrane and take part in fat-related or signaling pathways. Lastly, the Trypsin-like serine protease domain includes cleavage and catalytic motifs that aligns closely with known proteases like Prothrombin and Acrosin, both of which mediate cleavage reactions such as blood clotting and fertilization. These functional predictions and correlations collectively suggest that the protein is involved in mitochondrial apoptosis, lipid metabolism, membrane association, and protein degradation, supporting its multifaceted role in cell death, stress response, and possibly immune regulation.

b. Correlation between functional domains, membrane domains and secondary structure

All four secondary structure prediction tools—PSIPred, JPred4, PREDATOR, and SCRATCH—indicate that the protein consists of combination of alpha helices, beta strands, and coil regions. A strong alpha-helical region is predicted at the N-terminus (~10-269) aligning precisely with the Smac_DIABLO domain. This domain is known to adopt a helical bundle structure which is critical for its role in apoptotic signaling.⁷⁸ The Lipase domain (~271-605) displays a mixed pattern of alpha helices and mostly beta strands. Lipase enzymes classically adopt an alpha/beta-hydrolase fold⁷⁹ This

explains the dense and altering alpha and beta structures predicted in this region. The prediction by the secondary structure prediction experiments also suggest a well-organized catalytic core, suggesting for substrate binding and enzymatic activity. The PLAT domain is found in a variety of membrane or lipid associated protein.⁸⁰ The PLAT domain (~608-718) implicated in membrane association, is characterized by beta sandwich architecture composed mostly of parallel beta strands.⁸¹ Consistent with this, all predictors especially SCRATCH and PREDATOR– identify a dominant beta-stand signature with some unstructured regions across this region. This structural arrangement facilitates membrane docking through flat hydrophobic surfaces, which explains the biological role of PLAT domain in lipid interactions. Toward the C-terminus, the Trypsin-like serine protease domain (Tryp_Spc) (~719–954) is predicted to be even more heavily composed of extended beta strands. Classical serine proteases such as Trypsin and Elastase are known to form a conserved two-beta-barrel fold, and this is precisely the pattern seen across PSIPred, JPred4, PREDATOR, and SCRATCH, with long stretches of predicted beta strands and few helices.⁸² Altogether, these secondary structure predictions are highly correlated with known functional and membrane domains.

c. Correlation between functional domains, membrane domains and hydrophobicity

The functional domains in our protein along with its hydrophobicity are closely related to one another along with their overall role in protein function. In addition, the unanimous decision of the novel protein lacking any trans-membrane domains also plays into its predicted function. At the N-terminus of the protein is located at the Smac_DIABLO domain, which is associated with mitochondrial targeting and apoptotic signaling, along with its hydrophobic properties to deter IAP proteins (Inhibitor of apoptosis-binding protein), while also making the mitochondrial membrane more permeable.⁶² The PLAT_PL domain is located next to it, and uses hydrophobic surfaces to associate itself with the lipid bilayer.⁶³ From the three tools, the average hydrophobicity of the residues near the lipase domain are mostly hydrophobic, which matches its properties of breaking down the cell's lipid bilayer.⁶⁴ Tryp_SPC protease domain is located near the C-terminus, and is on average mostly hydrophobic according to the tools used in this project, which may reflect its role to cleave peptide bonds in proteins.⁶⁵ The absence of transmembrane domains allows the protein to remain mobile while it is associating with lipid surfaces.

d. Subcellular Localization Prediction

This protein is predicted to have two localizations. The protein likely travels first to the mitochondria, since a Smac_DIABLO domain was detected along with a mitochondrial targeting signal. After, it is most likely secreted into the extracellular space, since a targeting sequence to this location was also predicted. Although it is not a transmembrane protein, it could associate superficially with mitochondrial intermembrane using its

scattered hydrophobic patches. The Lipase domain would also be positioned near membrane surfaces, so that it can hydrolyze lipids. The PLAT_PL domain is located next to the lipase domain, which could increase binding specificity. Finally, the Tryp_SPc domain is most likely active near the cytoplasmic side of the membrane so it can degrade proteins associating with lipid bilayers. The scattered hydrophobicity matches with its binding role, and also supports the prediction that it is not a fixed membrane bound protein. Subcellular localization prediction showing mitochondrial and extracellular activity makes sense given the domain functions. This protein most likely helps destabilize mitochondrial membranes and degrade its proteins during apoptosis or cellular damage.

- e. **Function and molecular mechanism of action:** The protein is first thought to be brought to mitochondria due to its N-terminal signalling sequence and SMAC_Diablo domain. After entering the cell, then the mitochondria, it most likely prevents IAPs from blocking caspase activation, allowing apoptosis to proceed. PLAT_PL domain is most likely associated with the surface of the internal mitochondrial membrane, since the β -sandwich surrounding it makes it easier to do so. Moreover, the lack of transmembrane domains suggests that it interacts superficially with the mitochondrial membrane. The Catalytic triad in the lipase domain then could cleave the Try_SPc domain, causing it to go through conformational change and activation.⁸⁶ Active form of Try_SPc degrades membrane proteins while the Lipase domain breaks down mitochondrial membranes. The Calcium ligand most likely plays a role in this process, since it is crucial for stability and signalling. After degradation, the protein is released after apoptosis and guided into the extracellular region through the signaling peptide.

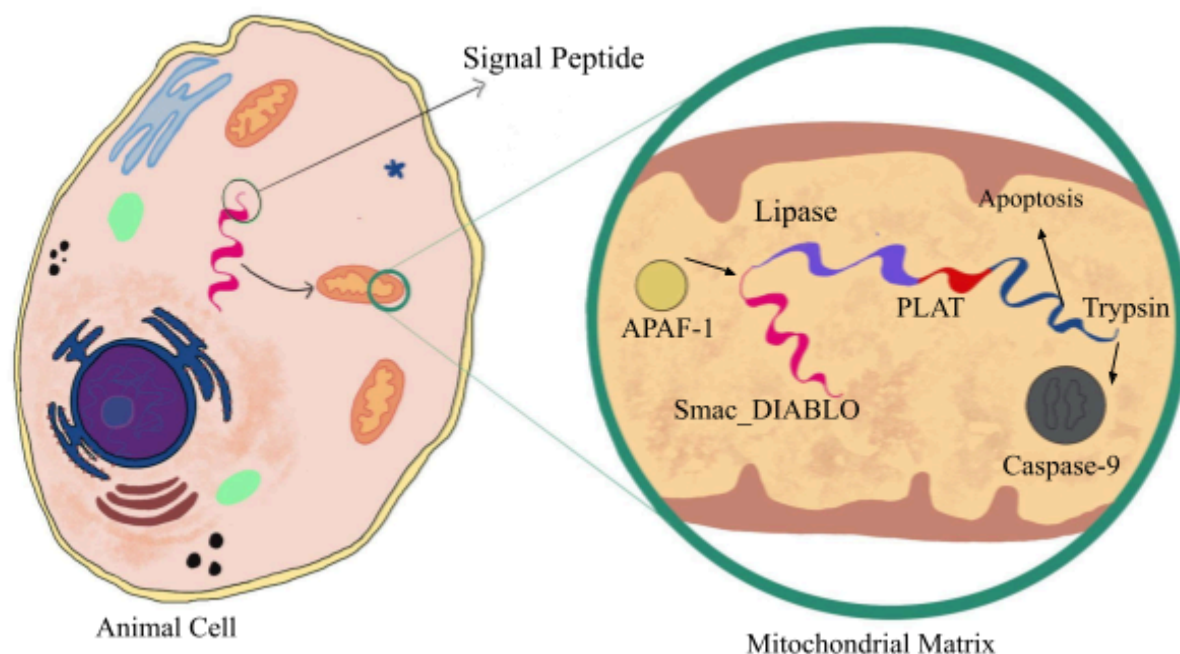
f. **Diagram:**

Figure 6.1. The diagram shows the molecular mechanism of the protein within the animal cell. The protein is guided to the mitochondria via its signal peptide. The Smac_DIABLO domain interacts with APAF-1, promoting apoptosome formation. The apoptosome recruits and activates procaspase-9, which in turn activates executioner caspases like caspase-9, leading to apoptosis.

- g. Cell Type:** The novel protein, Mi-lipotrymase, is most likely expressed primarily in the pancreatic cells. The protein's composition of Smac_DIABLO, lipase, PLAT_PL, and trypsin-like protease (Tryp_SPc) domains suggests that it has an important role in apoptotic activity. First, The presence of the Smac_DIABLO domain implies it may promote apoptosis. This aligns with the pancreas's need to regulate cell death to prevent enzyme leakage and tissue damage, as seen in conditions like pancreatitis. The other functional domains are also important. The lipase and PLAT_PL domains suggest membrane-associated lipid cleavage of the mitochondrial membrane, while the Tryp_SPc domain indicates protease activity, potentially involved in degrading mitochondrial membrane proteins. It's ligands, like C8E which breaks down membranes, along with MUP, which activates enzymes, also justify the protein's role in apoptosis.
- h. Further Predictions:** While the protein plays a major role in apoptosis, it may also function as a regular lipase and protease. Since it is predicted to be located in the pancreas, these two functions are implied. Its function as an important regulator for

apoptosis may only be activated under certain conditions or stressors. Therefore, potential mutations occurring in the lipase, PLAT_PL, and/or trypsin-like protease domains could cause potential diseases and ailments.

i. Disease(s) prediction

An important reason for analyzing protein sequences and determining their functions is to understand diseases and potentially find cures. By studying how proteins fold, interact with other molecules, and perform specific biological roles, we can identify where things go wrong in disease states. Mutations, misfolding, or loss of function in key proteins are often at the root of many disorders, from cancer to neurodegeneration.

- i. One disease that could be associated with this protein is cancer. The Smac/DIABLO domain promotes apoptosis by antagonizing Inhibitor of Apoptosis Proteins (IAPs), which normally block caspase activity and prevent cell death.⁸³ When Smac/DIABLO binds to IAPs, it allows apoptosis to happen properly. If this protein is overactive, it could cause too much apoptosis, leading to tissue damage. But if it's underactive, it would reduce apoptosis, meaning damaged or abnormal cells that should have died could survive and keep growing. This can give cancer cells an advantage by letting them avoid cell death and continue dividing. Many tumors have been shown to increase IAP levels to block apoptosis, and if Smac/DIABLO is not working well, this effect could become even stronger, promoting cancer development.
- ii. This protein could also be linked to metabolic diseases like obesity and type 2 diabetes. The Lipase domain plays a key role in lipid metabolism because it helps break down fats. Lipid remodeling is really important for how fat tissue expands and stores energy. If the lipase activity in this protein isn't working properly, it could disturb normal lipid breakdown, leading to fat accumulation where it shouldn't happen. It could also mess up mitochondrial energy production and disrupt insulin signaling pathways, both of which are major factors in metabolic diseases. So if this protein is dysfunctional, it might contribute to conditions like insulin resistance, abnormal fat storage, and overall problems with energy balance in the body.

j. Name of the protein:

The name of this protein is Mi-lipotrymase, a carefully constructed term that reflects both its structural domains and biological function. This name incorporates all the major domains of our novel protein. The Mi in the name stand for Mitochondria, this comes from the subcellular location of the protein, and the fact that one of the domains, Smac_DIABLO is involved in apoptosis. The middle component, "lipo", represents the presence of a lipase domain, indicating the protein's ability to interact with or break down

lipid membranes. The segment “try” references the trypsin-like serine protease domain (Tryp_Spc). Finally, the suffix “mase” reflects its overall enzymatic activity. Together, *Mi-lipotrymase* captures the integrated molecular function and domain architecture of this novel, apoptosis-associated protein.

k. Future Research

Research on protein sequences is an ongoing process that continues to expand as bioinformatics tools become more advanced and more data is collected. In the case of this protein, there are still properties that would benefit from further investigation to fully understand its biological role. Two key areas stand out that, if studied in more detail, could significantly help clarify the protein’s function and disease relevance.

One area that would benefit from further research is disease prediction based on the protein’s domains and motifs. Since this protein contains important functional regions like the Smac/DIABLO domain (apoptosis-related) and the Lipase domain (lipid metabolism-related), it would be valuable to predict diseases associated with mutations in these domains. These predictions could initially be made using bioinformatics tools based on known patterns in similar proteins, and then confirmed through wet lab experiments by introducing specific mutations and observing changes in apoptosis, lipid degradation, or membrane interactions. If mutations disrupt expected functions, this could validate links to diseases like cancer, obesity, or mitochondrial disorders.

Furthermore, studying the degradation pathway of the protein would provide insight into its functions before degradation and turnover in different cellular compartments. Protease protection assays could help determine if the protein is properly imported into the mitochondria or remains stuck on the mitochondrial surface. Pulse-chase experiments using labeled amino acids could track the synthesis, import, and degradation rates of the protein over time, giving a full view of its dynamic life cycle. Understanding these processes would be crucial, especially since mislocalization or impaired degradation of apoptosis-regulating proteins like those containing Smac/DIABLO domains can have serious consequences, such as promoting cancer or metabolic diseases. Therefore, future work should integrate both bioinformatics predictions and rigorous experimental validation to fully uncover the role of this protein at each stage of its life within the cell.

Credits

Ritika: 50%

- 1) Abstract
- 2) Objectives section
- 3) Background
- 4) Tools and methods:
 - a) Functional/structural domains and motifs, sequence motifs, and secondary structures
 - b) Principle of algorithms:
 - i) Functional/structural domains and motifs, sequence motifs, and secondary structures
- 5) Results:
 - a) Functional/structural domains and motifs, sequence motifs, and secondary structures
 - b) 3D Structures of known domains
 - c) 3D structure of protein: I-TASSER
 - d) Rationale of using I-TASSER and Alpha Fold
- 6) Discussions and Conclusions:
 - a) Correlation between previously well-established unique, identifying features, and correlation between functional domains, membrane domains and secondary structure
 - b) Disease(s) prediction
 - c) Function and molecular mechanism of action
 - d) Future Research

Andrew: 50%

- 1) Background
- 2) Tools and methods:
 - a) Hydrophilic/hydrophobic Regions, trans-membrane domains, subcellular localization/protein-sorting signals, and function predictions
 - b) Principle of algorithms:
 - i) Hydrophilic/hydrophobic regions, subcellular localization/protein-sorting signals, and function predictions
- 3) Results:
 - a) Hydrophilic/hydrophobic regions, subcellular localization/protein-sorting signals, and function predictions
 - b) Domains table
 - c) Composite diagram
 - d) 3D structure of protein: AlphaFold
- 4) Discussions and Conclusions:
 - a) Correlation between functional domains, membrane domains and hydrophobicity
 - b) Subcellular localization prediction
 - c) Function and molecular mechanism of action

X_____

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References

1. National Center for Biotechnology Information (NCBI). *Conserved Domain Database (CDD)*. U.S. National Library of Medicine, Mar. 2024. Web. Accessed 25 Apr. 2025. <https://www.ncbi.nlm.nih.gov/cdd/>
2. EMBL-EBI. *InterPro: Protein Sequence Analysis & Classification*. European Bioinformatics Institute, 2024. Web. Accessed 25 Apr. 2025. <https://www.ebi.ac.uk/interpro/>
3. Letunic, Ivica, and Peer Bork. *SMART: Simple Modular Architecture Research Tool*. EMBL, 2024. Web. Accessed 25 Apr. 2025. <http://smart.embl-heidelberg.de/>
4. Eddy, Sean R. *HMMER: Biosequence Analysis Using Profile Hidden Markov Models*. Howard Hughes Medical Institute, 2024. Web. Accessed 25 Apr. 2025. <http://hmmer.org/>
5. Bagheri, Hamed, and Lukasz Kurgan. *NsitePred: Predicting Nucleotide-Binding Residues from Protein Sequences*. Virginia Commonwealth University, 2024. Web. Accessed 25 Apr. 2025. <https://biomine.cs.vcu.edu/servers/NsitePred/>
6. Yang, Jianyi, and Yang Zhang. *COACH: A Meta Server for Protein–Ligand Binding Site Prediction*. Zhang Lab, University of Michigan, 2024. Web. Accessed 25 Apr. 2025. <https://zhanggroup.org/COACH/>
7. Waterhouse, Andrew M., et al. *SWISS-MODEL: Homology Modelling of Protein Structures and Complexes*. Swiss Institute of Bioinformatics, 2024. Web. Accessed 25 Apr. 2025. <https://swissmodel.expasy.org/>
8. Paysan-Lafosse, Thomas, et al. "InterPro in 2024: improving protein sequence annotations by extending coverage across the tree of life." *Nucleic Acids Research*, vol. 52, no. W1, 2024, pp. W287–W294. Oxford University Press. Web. Accessed 25 Apr. 2025. <https://academic.oup.com/nar/article/52/W1/W287/7673487>
9. Buchan, Daniel W. A., and David T. Jones. *PSIPRED: Protein Structure Prediction Server*. UCL Bioinformatics Group, 2024. Web. Accessed 25 Apr. 2025. <https://bio.tools/psipred>
10. Drozdetskiy, A., Cole, C., Procter, J., & Barton, G. J. (2015). JPred4: a protein secondary structure prediction server. *Nucleic Acids Research*, 43(W1), W389–W394. <https://doi.org/10.1093/nar/gkv332>
11. Cole, Colin, et al. *JPred: Protein Secondary Structure Prediction Server*. University of Dundee, 2024. Web. Accessed 25 Apr. 2025. <https://www.compbio.dundee.ac.uk/jpred/>
12. Coleman, Andrew. *PREDATOR: Protein Secondary Structure Prediction Tool*. PRABI Lyon-Gerland, 2024. Web. Accessed 25 Apr. 2025. https://npsa-prabi.ibcp.fr/cgi-bin/npsa_automat.pl?page=/NPSA/npsa_predator.html
13. Cheng, Jianzhu, et al. *SCRATCH: Protein Structure and Structural Feature Prediction Server*. University of California, Irvine, 2024. Web. Accessed 25 Apr. 2025. <https://scratch.proteomics.ics.uci.edu/>

14. National Center for Biotechnology Information (NCBI). *Conserved Domain Database (CDD): News and Updates*. U.S. National Library of Medicine, 2024. Web. Accessed 25 Apr. 2025. https://www.ncbi.nlm.nih.gov/Structure/cdd/docs/cdd_news.html
15. Marchler-Bauer, A., et al. "CDD: Specific Functional Annotation with the Conserved Domain Database." *Nucleic Acids Research*, vol. 37, no. Database, Nov. 2008, pp. D205–10. <https://doi.org/10.1093/nar/gkn845>.
16. Marchler-Bauer, Aron, et al. "CDD/SPARCLE: functional classification of proteins via subfamily domain architectures." *Nucleic Acids Research*, vol. 45, no. D1, 2017, pp. D200–D203. Web. Accessed 25 Apr. 2025. <http://academic.oup.com/nar/article/45/D1/D200/2605748?login=true>
17. Blum, Matthias, et al. "InterPro: the protein sequence classification resource in 2025." *Nucleic Acids Research*, vol. 53, no. D1, 6 Jan. 2025, pp. D444–D456. <https://doi.org/10.1093/nar/gkae1082>. Accessed 25 Apr. 2025.
18. Eddy, S. R. "Profile hidden Markov models." *Bioinformatics*, vol. 14, no. 9, 1998, pp. 755–763. <https://doi.org/10.1093/bioinformatics/14.9.755>. Accessed 25 Apr. 2025.
19. Blum, Matthias, et al. "The InterPro protein families and domains database: 20 years on." *Nucleic Acids Research*, vol. 49, no. D1, 8 Jan. 2021, pp. D344–D354. <https://doi.org/10.1093/nar/gkaa977>. Accessed 25 Apr. 2025.
20. Schultz, J., F. Milpetz, P. Bork, and C. P. Ponting. "SMART, a simple modular architecture research tool: identification of signaling domains." *Proceedings of the National Academy of Sciences of the United States of America*, vol. 95, no. 11, 1998, pp. 5857–5864. <https://doi.org/10.1073/pnas.95.11.5857>. Accessed 25 Apr. 2025.
21. Letunic, Ivica, Supriya Khedkar, and Peer Bork. "SMART: recent updates, new developments and status in 2020." *Nucleic Acids Research*, vol. 49, no. D1, 8 Jan. 2021, pp. D458–D460. <https://doi.org/10.1093/nar/gkaa937>. Accessed 25 Apr. 2025.
22. Schultz, Jörg, et al. "SMART: a web-based tool for the study of genetically mobile domains." *Nucleic Acids Research*, vol. 28, no. 1, Jan. 2000, pp. 231–234. <https://doi.org/10.1093/nar/28.1.231>. Accessed 25 Apr. 2025.
23. Potter, Simon C., et al. "HMMER web server: 2018 update." *Nucleic Acids Research*, vol. 46, Web Server issue, 14 June 2018, pp. W200–W204. <https://doi.org/10.1093/nar/gky448>. Accessed 25 Apr. 2025.
24. HMMER Web Server Team. *Searches – HMMER Documentation*. HMMER Web Server, 2024. Web. Accessed 25 Apr. 2025. <https://hmmer-web-docs.readthedocs.io/en/latest/searches.html>
25. Finn, Robert D., Jody Clements, and Sean R. Eddy. "HMMER web server: interactive sequence similarity searching." *Nucleic Acids Research*, vol. 39, suppl_2, 1 July 2011, pp. W29–W37. <https://doi.org/10.1093/nar/gkr367>. Accessed 25 Apr. 2025.
26. Chen, Ke, Marcin J. Mizianty, and Lukasz Kurgan. "Prediction and analysis of nucleotide-binding residues using sequence and sequence-derived structural descriptors."

- Bioinformatics*, vol. 28, no. 3, 1 Feb. 2012, pp. 331–341.
<https://doi.org/10.1093/bioinformatics/btr657>. Accessed 25 Apr. 2025.
27. Schwede, Torsten, et al. "SWISS-MODEL: an automated protein homology-modeling server." *Nucleic Acids Research*, vol. 31, no. 13, 1 July 2003, pp. 3381–3385.
<https://doi.org/10.1093/nar/gkg520>. Accessed 25 Apr. 2025.
 28. Roy, Ambrish, and Yang Zhang. "COFACTOR: an accurate comparative algorithm for structure-based protein function annotation." *Nucleic Acids Research*, vol. 40, Web Server issue, 2012, pp. W471–W477. <https://doi.org/10.1093/nar/gks372>. Accessed 25 Apr. 2025.
 29. Brylinski, Michal, and Jeffrey Skolnick. "FINDSITE: a threading-based method for ligand-binding site prediction and functional annotation." *Proceedings of the National Academy of Sciences*, vol. 105, no. 1, 2008, pp. 129–134.
<https://doi.org/10.1073/pnas.0707684105>. Accessed 25 Apr. 2025.
 30. Yang, Jianyi, and Yang Zhang. "Protein–ligand binding site recognition using complementary binding-specific substructure comparison and sequence profile alignment." *Bioinformatics*, vol. 28, no. 12, 2012, pp. 1533–1540.
<https://doi.org/10.1093/bioinformatics/bts222>. Accessed 25 Apr. 2025.
 31. Capra, John A., and Mona Singh. "Predicting functionally important residues from sequence conservation." *Bioinformatics*, vol. 23, no. 15, 2007, pp. 1875–1882.
<https://doi.org/10.1093/bioinformatics/btm270>. Accessed 25 Apr. 2025.
 32. Yang, Jianyi, and Yang Zhang. "S-SITE: protein–ligand binding site prediction by threading-based structural alignment." *Bioinformatics*, vol. 27, no. 7, 2011, pp. 1009–1010. <https://doi.org/10.1093/bioinformatics/btr062>. Accessed 25 Apr. 2025.
 33. Wu, Qi, et al. "COACH-D: improved protein–ligand binding sites prediction with refined ligand-binding poses through molecular docking." *Nucleic Acids Research*, vol. 46, Web Server issue, 28 May 2018, pp. W438–W442. <https://doi.org/10.1093/nar/gky439>. Accessed 25 Apr. 2025.
 34. IBM. *Neural Networks*. IBM Think, 2024. Web. Accessed 25 Apr. 2025.
<https://www.ibm.com/think/topics/neural-networks>
 35. Schäffer, Alejandro A., et al. "Improving the accuracy of PSI-BLAST protein database searches with composition-based statistics and other refinements." *Nucleic Acids Research*, vol. 29, no. 14, 15 July 2001, pp. 2994–3005.
<https://doi.org/10.1093/nar/29.14.2994>. Accessed 25 Apr. 2025.
 36. National Institutes of Health (NIH). *Biowulf User Guides*. High-Performance Computing at NIH, 2024. Web. Accessed 25 Apr. 2025. https://hpc.nih.gov/docs/user_guides.html
 37. Amazon Web Services. *Cross-Validation – Amazon Machine Learning Developer Guide*. AWS Documentation, 2024. Web. Accessed 25 Apr. 2025.
<https://docs.aws.amazon.com/machine-learning/latest/dg/cross-validation.html>

38. Drozdetskiy, Alexey, et al. "JPred4: a protein secondary structure prediction server." *Nucleic Acids Research*, vol. 43, no. W1, 1 July 2015, pp. W389–W394.
<https://doi.org/10.1093/nar/gkv332>. Accessed 25 Apr. 2025.
39. Joosten, Robbie P., et al. "A series of PDB related databases for everyday needs." *Nucleic Acids Research*, vol. 39, suppl_1, 1 Jan. 2011, pp. D411–D419.
<https://doi.org/10.1093/nar/gkq1105>. Accessed 25 Apr. 2025.
40. Buchan, Daniel W. A., et al. "Deep learning for the PSIPRED Protein Analysis Workbench." *Nucleic Acids Research*, vol. 52, suppl_W1, 15 May 2024, pp. W287–W293. <https://doi.org/10.1093/nar/gkae328>. Accessed 25 Apr. 2025.
41. Cheng, Jianlin, and Pierre Baldi. *SCRATCH: A Protein Structure and Structural Feature Prediction Server*. University of California, Irvine.
<https://mupro.proteomics.ics.uci.edu/extra/scratch.pdf>. Accessed 25 Apr. 2025.
42. Zhang Lab. *I-TASSER: Protein Structure & Function Predictions*. University of Michigan. <https://zhanggroup.org/I-TASSER/>. Accessed 25 Apr. 2025.
43. Rogers, S., Wells, R., & Rechsteiner, M. (1986). Amino acid sequences common to rapidly degraded proteins: The pest hypothesis. *Science*, 234(4774), 364–368.
<https://doi.org/10.1126/science.2876518>
44. Jiménez-Bremont, J. F., Chávez-Martínez, A. I., Ortega-Amaro, M. A., Guerrero-González, M. L., Jasso-Robles, F. I., Maruri-López, I., Liu, J.-H., Gill, S. S., & Rodríguez-Kessler, M. (2022). Translational and post-translational regulation of polyamine metabolic enzymes in plants. *Journal of Biotechnology*, 344, 1–10.
<https://doi.org/10.1016/j.jbiotec.2021.12.004>
45. Schuster, M. K., & Grabner, M. (2002, March). *EMBOSS-epstfind*. Emboss.
<https://galaxy-iuc.github.io/emboss-5.0-docs/epstfind.html>
46. Stellwagen, R. H. (1992). Involvement of sequences near both amino and carboxyl termini in the rapid intracellular degradation of tyrosine aminotransferase. *Journal of Biological Chemistry*, 267(33), 23713–23721.
[https://doi.org/10.1016/s0021-9258\(18\)35896-4](https://doi.org/10.1016/s0021-9258(18)35896-4)
47. Gasteiger E., Hoogland C., Gattiker A., Duvaud S., Wilkins M.R., Appel R.D., Bairoch A.; *Protein Identification and Analysis Tools on the ExPASy Server*; (In) John M. Walker (ed): *The Proteomics Protocols Handbook*, Humana Press (2005).pp. 571-607
48. C.N. Magnan & P. Baldi (2014). *SSpro/ACCpro 5: almost perfect prediction of protein secondary structure and relative solvent accessibility using profiles, machine learning and structural similarity*. *Bioinformatics*, vol 30 (18), 2592-2597.
<https://doi.org/10.1093/bioinformatics/btu352>
49. Tsirigos, K. D., Peters, C., Shu, N., Käll, L., & Elofsson, A. (2015). The TOPCONS web server for consensus prediction of membrane protein topology and signal peptides. *Nucleic Acids Research*, 43(W1). <https://doi.org/10.1093/nar/gkv485>

50. Persson, B. & Argos, P. (1994) Prediction of transmembrane segments in proteins utilising multiple sequence alignments *J. Mol. Biol.* 237, 182-192. <https://doi.org/10.1006/jmbi.1994.1220>
51. Savojardo, C, Martelli, PL, Fariselli, P, Profiti, G, Casadio, R (2018) BUSCA: an integrative web server to predict subcellular localization of proteins. *Nucleic Acids Research*, 46(W1), W459-W466. <https://doi.org/10.1093/nar/gky320>.
52. Hallgren, J., Tsirigos, K. D., Pedersen, M. D., Almagro Armenteros, J. J., Marcatili, P., Nielsen, H., Krogh, A., & Winther, O. (2022). *DeepTMHMM Predicts Alpha and Beta Transmembrane Proteins Using Deep Neural Networks*. <https://doi.org/10.1101/2022.04.08.487609>
53. Horton, P., Park, K. J., Obayashi, T., Fujita, N., Harada, H., Adams-Collier, C. J., & Nakai, K. (2007). WoLF PSORT: protein localization predictor. *Nucleic acids research*, 35(Web Server issue), W585–W587. <https://doi.org/10.1093/nar/gkm259>
54. Ødum, M. T., Teufel, F., Thummuluri, V., Almagro Armenteros, J. J., Johansen, A. R., Winther, O., & Nielsen, H. (2024). DeepLoc 2.1: Multi-label membrane protein type prediction using protein language models. *Nucleic Acids Research*, 52(W1). <https://doi.org/10.1093/nar/gkae237>
55. Bannai, H., Tamada, Y., Maruyama, O., Nakai, K., & Miyano, S. (2002). Extensive feature detection of N-terminal protein sorting signals. *Bioinformatics*, 18(2), 298–305. <https://doi.org/10.1093/bioinformatics/18.2.298>
56. Yu, C., Chen, Y., Lu, C., & Hwang, J. (2006). Prediction of protein subcellular localization. *Proteins: Structure, Function, and Bioinformatics*, 64(3), 643–651. <https://doi.org/10.1002/prot.21018>
57. Bhasin, M. and Raghava, G.P.S. (2004) ESLpred: SVM Based Method for Subcellular Localization of Eukaryotic Proteins using Dipeptide Composition and PSI-BLAST. *Nucleic Acids Research* 32:W414-9. <https://doi.org/10.1093/nar/gkh350>
58. Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., Tunyasuvunakool, K., Bates, R., Žídek, A., Potapenko, A., Bridgland, A., Meyer, C., Kohl, S. A., Ballard, A. J., Cowie, A., Romera-Paredes, B., Nikolov, S., Jain, R., Adler, J., ... Hassabis, D. (2021). Highly accurate protein structure prediction with alphafold. *Nature*, 596(7873), 583–589. <https://doi.org/10.1038/s41586-021-03819-2>
59. Almagro Armenteros, J. J., Salvatore, M., Emanuelsson, O., Winther, O., von Heijne, G., Elofsson, A., & Nielsen, H. (2019). Detecting sequence signals in targeting peptides using deep learning. *Life Science Alliance*, 2(5). <https://doi.org/10.26508/lsa.201900429>
60. Rainbow. (n.d.). *ChimeraX*. https://www.cgl.ucsf.edu/chimera/docs/UsersGuide/midas/rainbow.html?utm_source=chatgpt.com
61. Google DeepMind. (n.d.). Alphafold server. <https://alphafoldserver.com/welcome>

62. Lv, Z., Song, X., Xu, J., Jia, Z., Yang, B., Jia, Y., Qiu, L., Wang, L., & Song, L. (2019). The modulation of SMAC/Diablo on mitochondrial apoptosis induced by LPS in *Crassostrea gigas*. *Fish & Shellfish Immunology*, 84, 587–598.
<https://doi.org/10.1016/j.fsi.2018.10.035>
63. Kulke, M., Kurtz, E., Boren, D. M., Olson, D. M., Koenig, A. M., Hoffmann-Benning, S., & Vermaas, J. V. (2024). Plat Domain Protein 1 (Plat1/plafp) binds to the *Arabidopsis thaliana* plasma membrane and inserts a lipid. *Plant Science*, 338, 111900.
<https://doi.org/10.1016/j.plantsci.2023.111900>
64. Khan, F. I., Lan, D., Durrani, R., Huan, W., Zhao, Z., & Wang, Y. (2017). The LID domain in lipases: Structural and functional determinant of enzymatic properties. *Frontiers in Bioengineering and Biotechnology*, 5.
<https://doi.org/10.3389/fbioe.2017.00016>
65. U.S. National Library of Medicine. (n.d.). *CDD conserved protein domain family: Tryp_spc*. National Center for Biotechnology Information.
<https://www.ncbi.nlm.nih.gov/Structure/cdd/cd00190>
66. Chou, K. C., & Shen, H. B. (2008). ProtIdent: a web server for identifying proteases and their types by fusing functional domain and sequential evolution information. *Biochemical and biophysical research communications*, 376(2), 321–325.
<https://doi.org/10.1016/j.bbrc.2008.08.125>
67. Kulmanov, M., Zhapa-Camacho, F., & Hoehndorf, R. (2021). DeepGOWeb: Fast and accurate protein function prediction on the (semantic) web. *Nucleic Acids Research*, 49(W1). <https://doi.org/10.1093/nar/gkab373>
68. Udatha, D. B., Madsen, K. M., Panagiotou, G., & Olsson, L. (2015). Multiple nucleophilic elbows leading to multiple active sites in a single module esterase from *Sorangium cellulosum*. *Journal of structural biology*, 190(3), 314–327.
<https://doi.org/10.1016/j.jsb.2015.04.009>
69. Bateman, A., & Sandford, R. (1999). The Plat Domain: A new piece in the PKD1 puzzle. *Current Biology*, 9(16). [https://doi.org/10.1016/s0960-9822\(99\)80380-7](https://doi.org/10.1016/s0960-9822(99)80380-7)
70. U.S. National Library of Medicine. (2025, April 9). *RecName: Full=serine protease HTRA2, mitochondrial; AltName: Full=high - protein - NCBI*. National Center for Biotechnology Information. <https://www.ncbi.nlm.nih.gov/protein/Q9VFJ3.1>
71. U.S. National Library of Medicine. (2010, November 12). *Mitochondrial arts protein [homo sapiens] - protein - NCBI*. National Center for Biotechnology Information.
<https://www.ncbi.nlm.nih.gov/protein/AAG45673.1>
72. Jenkins, G. (2007). The role of proteases in transforming growth factor- β activation. *The International Journal of Biochemistry & Cell Biology*, 40(6–7), 1068–1078.
<https://doi.org/10.1016/j.biocel.2007.11.026>

73. ScienceDirect Topics. (n.d.). *Chymotrypsin*. Chymotrypsin - an overview .
<https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/chymotrypsin>
74. U.S. National Library of Medicine. (n.d.). *ALOX5 arachidonate 5-lipoxygenase [Homo Sapiens (human)] - gene - NCBI*. National Center for Biotechnology Information.
<https://www.ncbi.nlm.nih.gov/gene/240>
75. U.S. National Library of Medicine. (n.d.-b). *Triglyceride lipase [Anguilla japonica] - protein - NCBI*. National Center for Biotechnology Information.
<https://www.ncbi.nlm.nih.gov/protein/19032297?report=GenPept>
76. U.S. National Library of Medicine. (2011, July 17). *Chain K, prethrombin-2 - protein - NCBI*. National Center for Biotechnology Information.
https://www.ncbi.nlm.nih.gov/protein/1MKX_K?report=GenPept
77. U.S. National Library of Medicine. (2025, April 9). *ACR acrosin [Homo Sapiens (human)] - gene - NCBI*. National Center for Biotechnology Information.
<https://www.ncbi.nlm.nih.gov/gene?Db=gene&Cmd=DetailsSearch&Term=49>
78. Chai, J., Du, C., Wu, J. W., Kyin, S., Wang, X., and Shi, Y. "Structural and biochemical basis of apoptotic activation by Smac/DIABLO." *Nature*, vol. 406, no. 6798, 2000, pp. 855–862. <https://doi.org/10.1038/35022514>. Accessed 29 Apr. 2025.
79. Ollis, D. L., et al. "The alpha/beta hydrolase fold." *Protein Engineering*, vol. 5, no. 3, Apr. 1992, pp. 197–211. <https://doi.org/10.1093/protein/5.3.197>. Accessed 29 Apr. 2025.
80. "Conserved Protein Domain Family: PLAT." *National Center for Biotechnology Information*, <https://www.ncbi.nlm.nih.gov/Structure/cdd/cddsrv.cgi?uid=216521>. Accessed 29 Apr. 2025.
81. Bateman, Alex, and Richard Sandford. "The PLAT domain: a new piece in the PKD1 puzzle." *Current Biology*, vol. 9, no. 16, 26 Aug. 1999, pp. R588–R590.
[https://doi.org/10.1016/S0960-9822\(99\)80380-7](https://doi.org/10.1016/S0960-9822(99)80380-7). Accessed 29 Apr. 2025.
82. Perona, J. J., and C. S. Craik. "Structural basis of substrate specificity in the serine proteases." *Protein Science*, vol. 4, no. 3, Mar. 1995, pp. 337–360.
<https://doi.org/10.1002/pro.5560040301>. Accessed 29 Apr. 2025.
83. Du, C., M. Fang, Y. Li, L. Li, and X. Wang. "Smac, a mitochondrial protein that promotes cytochrome c–dependent caspase activation by eliminating IAP inhibition." *Cell*, vol. 102, no. 1, 7 July 2000, pp. 33–42. [https://doi.org/10.1016/s0092-8674\(00\)00008-8](https://doi.org/10.1016/s0092-8674(00)00008-8). Accessed 29 Apr. 2025.
84. Söding, J., Biegert, A., & Lupas, A. N. (2005). The HHpred interactive server for protein homology detection and structure prediction. *Nucleic acids research*, 33(Web Server issue), W244–W248. <https://doi.org/10.1093/nar/gki408>
85. U.S. National Library of Medicine. (n.d.-c). *CDD conserved protein domain family: PLAT_LPL*. National Center for Biotechnology Information.
<https://www.ncbi.nlm.nih.gov/Structure/cdd/cddsrv.cgi?uid=cd01758>

86. Khan, F. I., Lan, D., Durrani, R., Huan, W., Zhao, Z., & Wang, Y. (2017). The Lid Domain in Lipases: Structural and Functional Determinant of Enzymatic Properties. *Frontiers in bioengineering and biotechnology*, 5, 16.
<https://doi.org/10.3389/fbioe.2017.00016>

Appendix